nucleoprotein in its native form, the conformation of the antigen as presented to the T-cell must be quite different.

In parallel, an entirely comparable series of experiments has established the role of MHC class II molecules in antigen presentation to helper T-cells. Initially, it was shown by Shevach and Rosenthal that lymphocyte proliferation to antigen *in vitro* could be blocked by antisera raised between two strains of guinea-pig which would have included antibodies to the MHC of the responding lymphocytes. More stringent evidence comes from the type of experiment in which a T-cell clone proliferating in response to ovalbumin on antigen-presenting cells with the H-$2A^b$ phenotype fails to respond if antigen is presented in the context of H-$2A^k$. However, if the H-$2A^k$ antigen-presenting cells are transfected with the genes encoding H-$2A^b$, they now communicate effectively with the T-cells (figure M5.1.2).

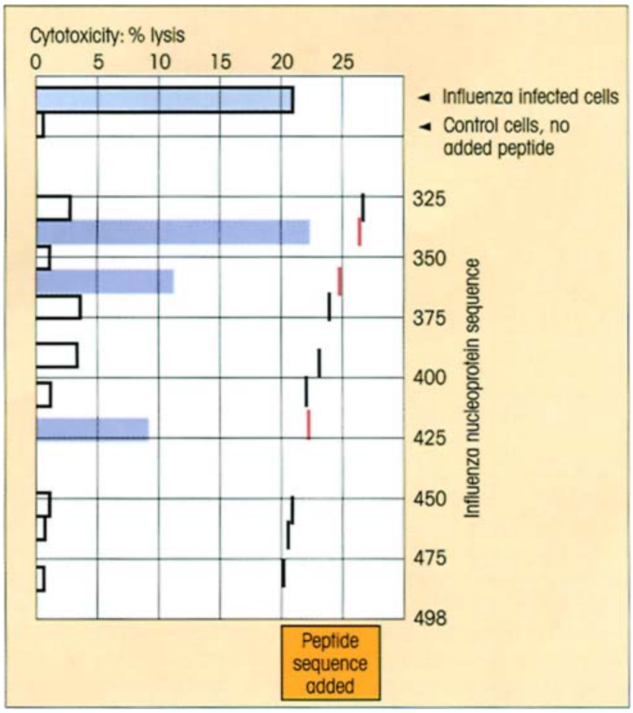

Figure 5.15. Cytotoxic T-cells, from a human donor, kill uninfected target cells in the presence of short influenza nucleoprotein peptides. The peptides indicated were added to ^{51}Cr-labeled syngeneic (i.e. same as T-cell donor) mitogen-activated lymphoblasts and cytotoxicity was assessed by ^{51}Cr release with a killer to target ratio of 50 : 1. The three peptides indicated in red induced good killing. Blasts infected with influenza virus of a different strain served as a positive control. (Reproduced from Townsend A.R.M. *et al.* (1986) *Cell* **44**, 959, with permission. Copyright ©1986 by Cell Press.)

class I and class II outermost domains (figure 4.11). Just how does it get there?

PROCESSING OF INTRACELLULAR ANTIGEN FOR PRESENTATION BY CLASS I MHC

Within the cytosol lurk proteolytic structures, the proteasomes, involved in the routine turnover and cellular degradation of proteins (figure 5.16). Cytosolic proteins destined for antigen presentation, including viral proteins, are degraded to peptides via a pathway involving these structures, although other cytosolic proteases including leucine- and aspartyl-aminopeptidases may also contribute to this antigen processing. In addition to cytosol-resident proteins, a proportion of membrane-bound and secretory proteins are transported from the endoplasmic reticulum (ER) back into the cytosol by the SEC61 molecular complex, and such proteins can then also undergo processing for class I presentation, as can proteins derived from mitochondria. Prior to processing, proteins are covalently linked to several ubiquitin molecules in an ATP-dependent process. The polyubiquitination targets the polypeptides to the proteasome (figure 5.16).

Only about 10% of the peptides produced by proteasomes are the optimal length (octamers or nonamers) to fit into the MHC class I groove; about 70% are likely to be too small to function in antigen presentation; and the remaining 20% would require further trimming by, for example, cytosolic aminopeptidases. The cytokine IFNγ increases the production of three catalytic proteosomal subunits, the polymorphic *l*ow *m*olecular weight *p*roteins LMP2 and LMP7, and the nonpolymorphic LMP10. These molecules replace the homologous catalytic subunits (β_1, β_5 and β_2, respectively) in the housekeeping proteasome to produce what has been termed the **immunoproteasome**, a process thought to modify the cleavage specificity in order to tailor peptide production for class I binding.

Both proteasome- and immunoproteasome-generated peptides are translocated into the ER by the *t*ransporters associated with *a*ntigen *p*rocessing (TAP1 and TAP2) (figure 5.17), a process which might also involve heat-shock protein family members. The newly synthesized class I heavy chain is retained in the ER by the molecular chaperone calnexin which is thought to assist in folding, disulfide bond formation and

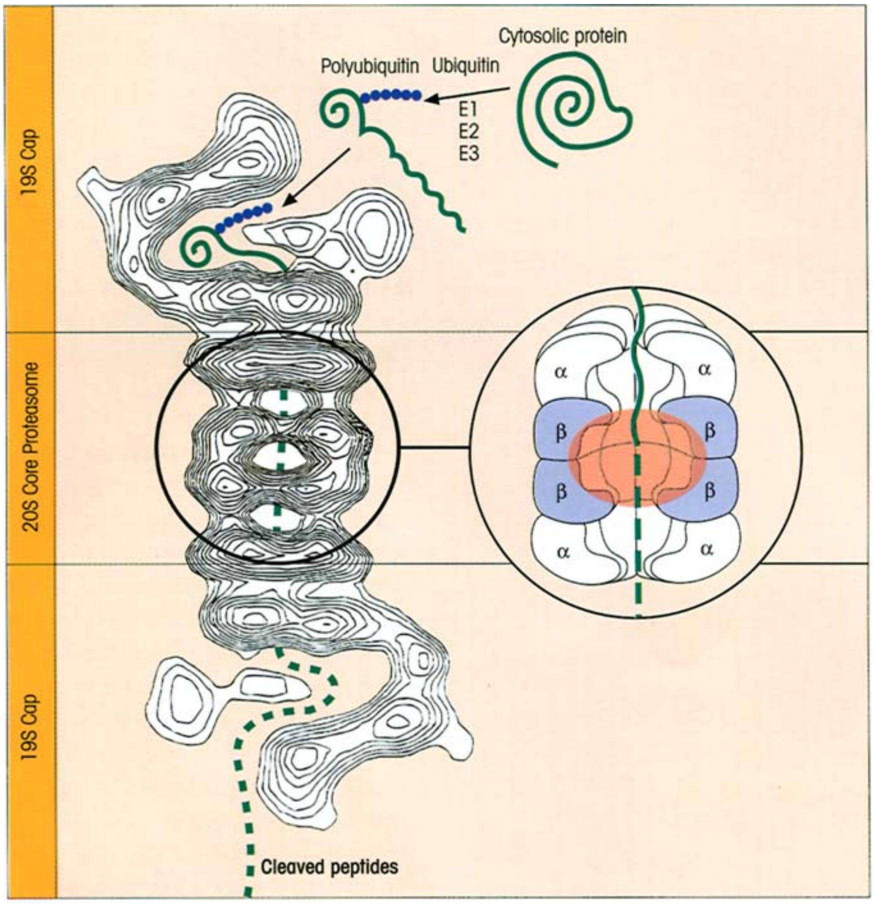

Figure 5.16. Cleavage of cytosolic proteins by the proteasome. Cytosolic proteins become polyubiquitinated in an ATP-dependent reaction in which the enzyme E1 forms a thiolester with the C-terminus of ubiquitin and then transfers the ubiquitin to one of 10–15 different E2 ubiquitin-carrier proteins. The C-terminus of the ubiquitin is then conjugated by one of a dozen or so E3 ubiquitin-protein ligase enzymes to a lysine residue on the polypeptide. There is specificity in these processes in that the individual E2 and E3 enzymes have preferences for different proteins. The ubiquitinated cytosolic protein binds to the ATPase-containing cap where ATP drives the unfolded protein chain through the hydrophobic conducting channels of the α-subunits into the central hydrolytic chamber where it is exposed to a variety of proteolytic activities associated with the different β-subunits. The whole 26S proteasome, which is a 2000 kDa complex of about 50 different subunits consisting of the 20S 650 kDa core proteasome with twin 19S 700 kDa regulatory caps, is displayed as a contour plot derived from electron microscopy and image analysis. The core proteasome is a cylindrical structure made up of 28 subunits arranged in four stacked rings. The cross-section of the proteasome reveals the site of proteolytic activity (red shading) within the two central rings, each comprising seven homologous but distinct β-subunits, three of which in each ring contain proteolytically active sites. The outer two α-rings are again each made up of seven different but homologous α-subunits, but these all lack proteolytic activity. A novel catalytic mechanism is involved in which the nucleophilic residue that attacks the peptide bonds is the hydroxyl group on the N-terminal threonine residue of the β-subunits. Three distinct peptidase activities have been associated with specific β-subunits. One is 'chymotrypsin-like' in that it hydrolyses peptides after large hydrophobic residues, one is 'trypsin-like' and cleaves after basic residues, and one hydrolyses after acidic residues. The LMP2, LMP7 and LMP10 (the latter often called MECL1; *m*ulticatalytic *e*ndopeptidase *c*omplex *l*ike-1) immunoproteasome-associated molecules show similar specificities but have enhanced chymotrypsin and trypsin activity and reduced postacidic cleavage compared to their counterparts in the housekeeping proteasome. (Based on Peters J.-M. *et al.* (1993) *Journal of Molecular Biology* **234**, 932 and Rubin D.M. & Finley D. (1995) *Current Biology* **5**, 854.)

promotion of assembly with β_2-microglobulin. In the human, but not in the mouse, calnexin is then replaced by calreticulin. The ER-resident protein, Erp57, which has thiol reductase, cysteine protease and chaperone functions, becomes associated with the complex of calreticulin–calnexin and class I heavy chain which now folds together with β_2-microglobulin. The empty class I molecule bound to these chaperones becomes linked to TAP1/2 by tapasin. Upon peptide loading, the class I molecule can dissociate from the various accessory molecules, and the now stable peptide–class I heavy chain–β_2-microglobulin complex traverses the Golgi stack and reaches the surface where it is a sitting target for the cytotoxic T-cell.

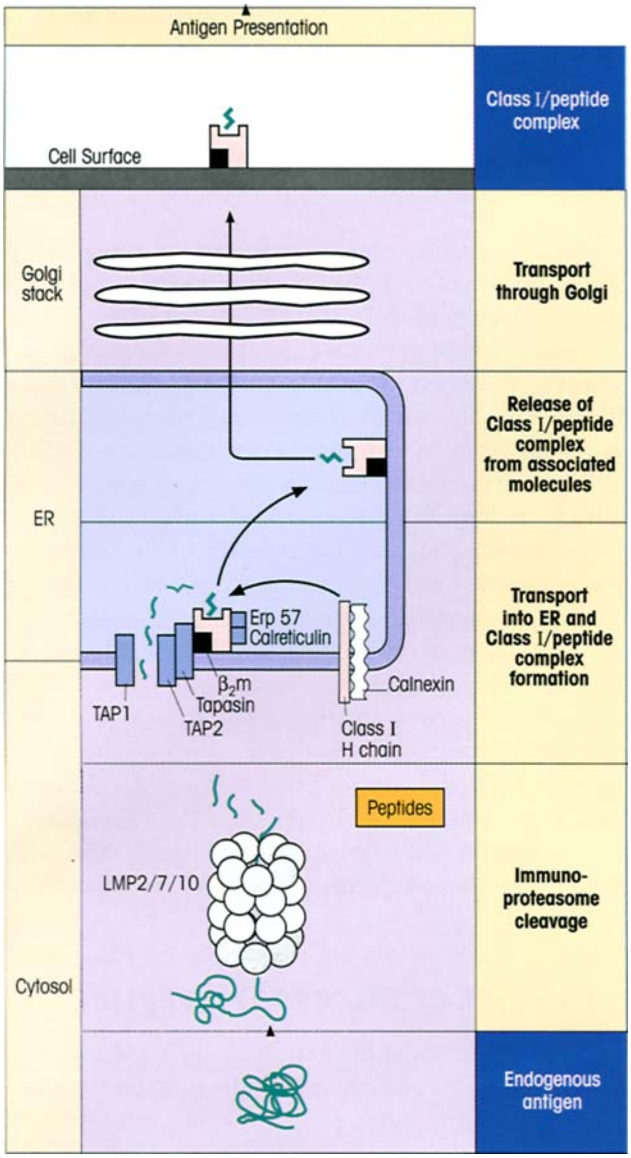

Figure 5.17. Processing and presentation of endogenous antigen by class I MHC. Cytosolic proteins are degraded by the proteasome complex into peptides which are transported into the endoplasmic reticulum (ER). TAP1 and TAP2 are members of the ABC family of ATP-dependent transport proteins and, under the influence of these transporters, the peptides are loaded into the groove of the membrane-bound class I MHC. The peptide–MHC complex is then released from all its associated transporters and chaperones, traverses the Golgi system, and appears on the cell surface ready for presentation to the T-cell receptor. Mutant cells deficient in TAP1/2 do not deliver peptides to class I and cannot function as cytotoxic T-cell targets. However, if they are transfected with a gene encoding the antigenic peptide linked to a cleavable signal sequence, the peptide is delivered to the ER without the need for TAP1/2 and the cells once again can become targets.

PROCESSING OF ANTIGEN FOR CLASS II MHC PRESENTATION FOLLOWS A DIFFERENT PATHWAY

Class II MHC complexes with antigenic peptide are generated by a fundamentally different intracellular mechanism, since the antigen-presenting cells which interact with T-helper cells need to sample the antigen from both the *extra*cellular and *intra*cellular compartments. In essence, a trans-Golgi vesicle containing class II has to intersect with a late endosome containing exogenous protein antigen taken into the cell by an endocytic mechanism.

Regarding the class II molecules themselves, these are assembled from α and β chains in the endoplasmic reticulum in association with the transmembrane **invariant chain (Ii)** (figure 5.18) which has several functions. Firstly, it acts as a dedicated chaperone to ensure correct folding of the nascent class II molecule. Secondly, an internal sequence of the luminal portion of Ii sits in the MHC groove to inhibit the precocious binding of peptides in the ER before the class II molecule reaches the endocytic compartment containing antigen. Additionally, combination of Ii with the αβ class II heterodimer inactivates a retention signal and allows transport to the Golgi. Finally, targeting motifs in the N-terminal cytoplasmic region of Ii ensure delivery of the class II-containing vesicle to the endocytic pathway.

Meanwhile, exogenous protein is taken up by endocytosis and, as the early endosome undergoes progressive acidification, is processed into peptides by endosomal proteases such as asparaginyl endopeptidase. The late endosomes, which ultimately mature into lysosomes, characteristically acquire *l*ysosomal-*a*ssociated *m*embrane *p*roteins (LAMPs), although the function of these molecules is still unclear. These late endosomes fuse with the vacuole containing the class II–Ii complex. Under the acidic conditions within these *M*HC class *II*-enriched *c*ompartments (MIICs), proteases degrade Ii except for the part sitting in the MHC groove which, for the time being, remains there as a peptide referred to as CLIP (*cl*ass II-associated *i*nvariant chain *p*eptide). An MHC-related dimeric molecule, DM, then catalyses the removal of CLIP and keeps the groove open so that peptides generated in the endosome can be inserted (figure 5.19). This process may be assisted by members of the hsp70 family which promiscuously bind unfolded peptides. Initial peptide binding is determined by the concentration of the peptide and its on-rate, but DM may subsequently assist in the removal of lower affinity peptides to allow their replacement by high affinity peptides, i.e. act as a peptide editor permitting the incorporation of peptides with the most stable binding characteristics, namely those with a slow off-rate. Particularly in B-cells an additional MHC-related molecule, DO, associates with DM bound to class II and modifies its function in a pH-dependent fashion. Its effect may be to favor the presentation of antigens internalized via the BCR over those taken up by fluid phase endocytosis. The tetraspanin family member CD82 is also present in the MIIC, though its role is unclear at present. The class II–peptide complexes are eventually transported to the membrane for presentation to T-helper cells.

Thus, in general, endogenous protein antigens are processed for class I presentation whilst exogenous protein antigens are processed for class II presentation. Processing of antigens for class II presentation is not, however, confined to soluble proteins taken up from the exterior, but can also encompass microorganisms whose antigens reach the lysosomal structures, either after direct phagocytosis or prolonged intracellular cohabitation. Proteins and peptides within the ER are also potential clients for the class II groove and could also make the journey to the MIIC. Conversely, class I-restricted responses can be generated against exogenous antigens, a process sometimes referred to as **cross-priming**. This may occur either by a TAP-dependent pathway in phagocytic cells where proteins are transferred from the phagosome to the cytosol, or by endocytosis of cell surface MHC class I molecules which then arrive in the class II-enriched compartments where peptide exchange occurs with sequences derived from the endocytic processing pathway.

THE NATURE OF THE 'GROOVY' PEPTIDE

The MHC grooves impose some well-defined restrictions on the nature and length of the peptides they ac-

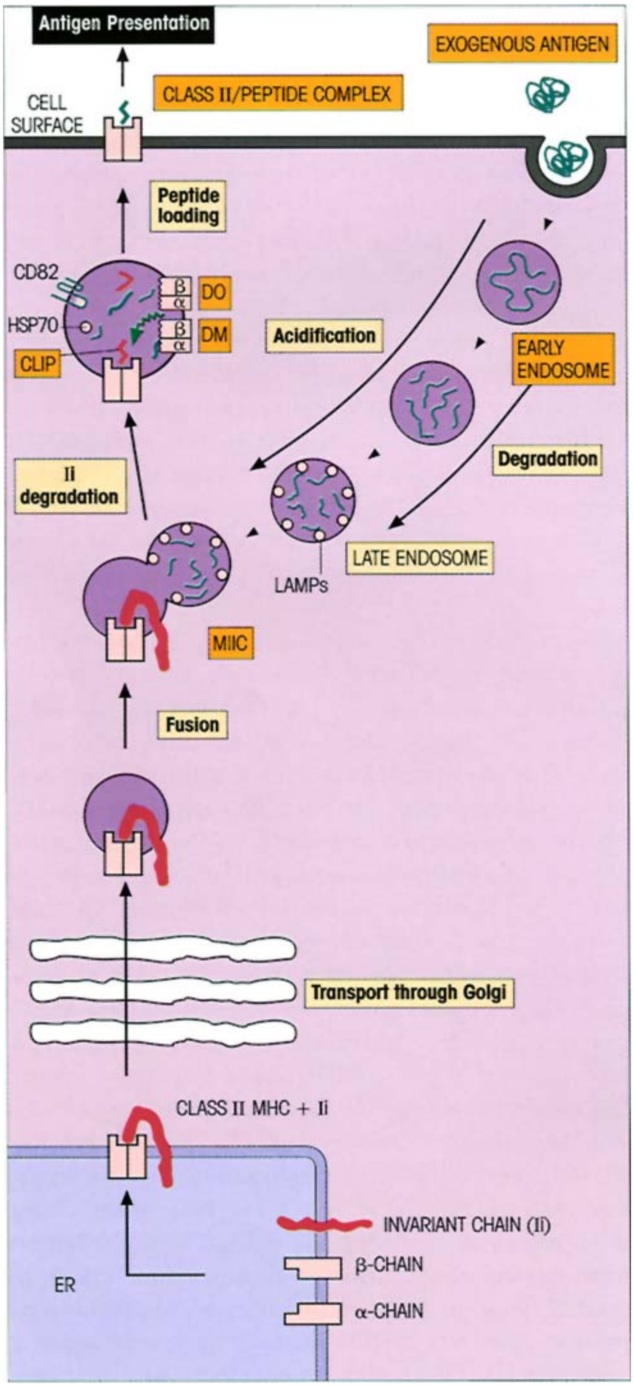

Figure 5.18. Processing and presentation of exogenous antigen by class II MHC. Class II molecules with Ii are assembled in the endoplasmic reticulum (ER) and transported through the Golgi to the trans-Golgi reticulum (actually as a nonamer consisting of three invariant, three α and three β chains—not shown). There it is sorted to a late endosomal vesicle with lysosomal characteristics known as MIIC (meaning MHC class II-enriched compartment) containing partially degraded protein derived from the endocytic uptake of exogenous antigen. Degradation of the invariant chain leaves the CLIP (class II-associated invariant chain peptide) lying in the groove but, under the influence of the DM molecule, this is replaced by other peptides in the vesicle including those derived from exogenous antigen, and the complexes are transported to the cell surface for presentation to T-helper cells. This version of events is supported by the finding of high concentrations of invariant chain CLIP associated with class II in the MIIC vacuoles of DM-deficient mutant mice which are poor presenters of antigen to T-cells.

Figure 5.19. MHC class II transport and peptide loading illustrated by Tulp's gently vulgar cartoon. (Reproduced from Benham A. *et al*. (1995) *Immunology Today* **16**, 361, with permission of the authors and Elsevier Science Ltd.)

commodate and the pattern varies with different MHC alleles. Otherwise, at the majority of positions in the peptide ligand, a surprising degree of redundancy is permitted and this relates in part to residues interacting with the T-cell receptor rather than the MHC.

Binding to MHC class I

X-ray analysis reveals the peptides to be tightly mounted along the length of the groove in an extended configuration with no breathing space for α-helical structures (figure 5.20). The N- and C-termini are tightly H bonded to conserved residues at each end of the groove, independently of the MHC allele.

The naturally occurring peptides can be extracted from purified MHC class I and sequenced. They are predominantly 8–9 residues long; longer peptides bulge upwards out of the cleft. Analysis of the peptide pool sequences usually gives strong amino acid signals at certain key positions (table 5.2). These are called **anchor positions** and represent the preferred amino acid side-chains which fit into allele-specific pockets in the MHC groove (figure 5.21a). There are usually two, sometimes three, such major anchor positions for class I-binding peptides, one at the C-terminal end and the other frequently at position 2 (P2), but it may also occur at P3, P5 or P7. Sometimes, a major anchor pocket may be replaced by two or three more weakly binding secondary binding pockets. Even with the constraints of two or three anchor motifs, each MHC class I allele can accommodate a considerable number of different peptides.

Except in the case of viral infection, the natural class I ligands will be self peptides derived from proteins endogenously synthesized by the cell, histones, heat-shock proteins, enzymes, leader signal sequences, and so on. It turns out that 75% or so of these peptides originate in the cytosol (figure 5.22) and most of them will be in low abundance, say 100–400 copies per cell. Thus proteins expressed with unusual abundance, such as oncofetal proteins in tumors and viral antigens in infected cells, should be readily detected by resting T-cells.

Binding to MHC class II

Unlike class I, where the allele-independent H bonding to the peptide is focused at the N- and C-termini, the class II groove residues H bond along the entire length of the peptide with links to the atoms forming the main chain. With respect to class II allele-specific binding pockets for peptide side-chains, motifs based on three or four major anchor residues seem to be the order of the day (figure 5.21b). Secondary binding pockets with less strict preference for individual side-

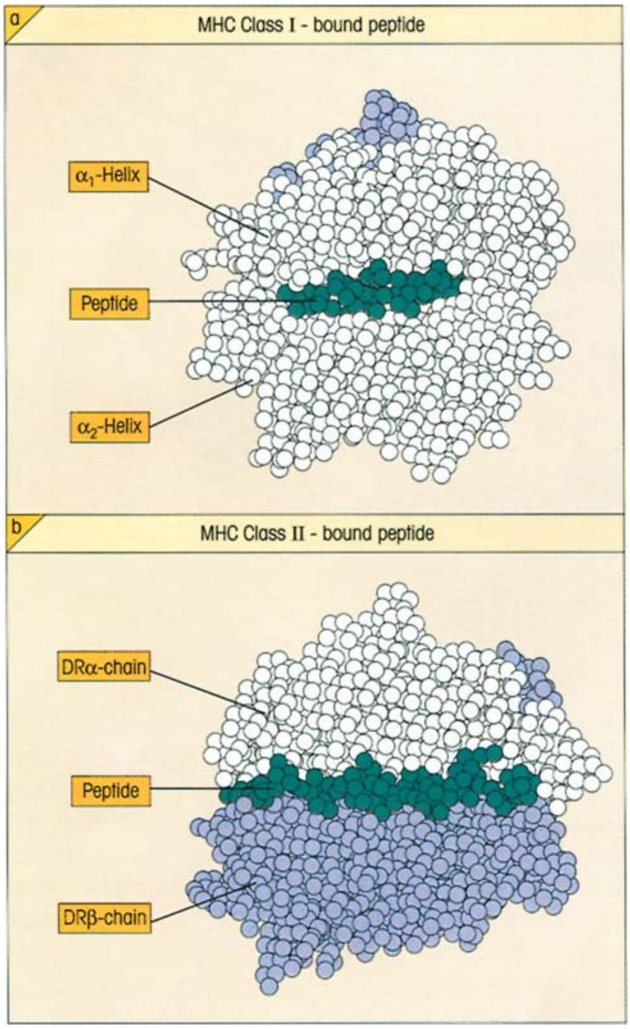

Figure 5.20. Binding of peptides to the MHC cleft. T-cell receptor 'view' looking down on the α-helices lining the cleft (cf. figure 4.11b) represented in space-filling models. (a) Peptide 309–317 from HIV-1 reverse transcriptase bound tightly within the class I HLA-A2 cleft. In general, one to four of the peptide side-chains point towards the TCR, giving a solvent accessibility of 17–27%. (b) Influenza hemagglutinin 306–318 lying in the class II HLA-DR1 cleft. In contrast with class I, the peptide extends out of both ends of the binding groove and from four to six out of an average of 13 side-chains point towards the TCR, increasing solvent accessibility to 35%. (Based on Vignali D.A.A. & Strominger J.L. (1994) *The Immunologist* **2**, 112, with permission of the authors and publisher.)

chains can still modify the affinity of the peptide–MHC complex, while 'nonpockets' may also influence preferences for particular peptide sequences, especially if steric hindrance becomes a factor. Unfortunately, we cannot establish these preferences for the individual residues within a given peptide because the open nature of the class II groove places no constraint on the length of the peptide, which can dangle nonchalantly from each end of the groove, quite unlike the straitjacket of the class I ligand site (figures 5.20 and 5.21). Thus, as noted earlier, each class II molecule binds a collection of peptides with a spectrum of lengths rang-

Table 5.2. Natural MHC class I peptide ligands contain two allele-specific anchor residues. (Based on Rammensee H.G., Friede T. & Stevanovic S. (1995) *Immunogenetics* **41**, 178.) Letters represent the Dayhoff code for amino acids; where more than one residue predominates at a given position, the alternative(s) is given; =any residue.

Class I allele	Amino acid position 1	2	3	4	5	6	7	8	9
H-2K^d	•	Y	•	•	•	•	•	•	I/L
H-2K^b	•	•	•	•	Y/F	•	•	L/M	
H-2D^b	•	•	•	•	N	•	•	•	L/M/I
HLA-A*0201	•	L/M/I	•	•	•	•	•	•	L/V/I/M
HLA-B*2705	•	R	•	•	•	•	•	•	R/K/L/F

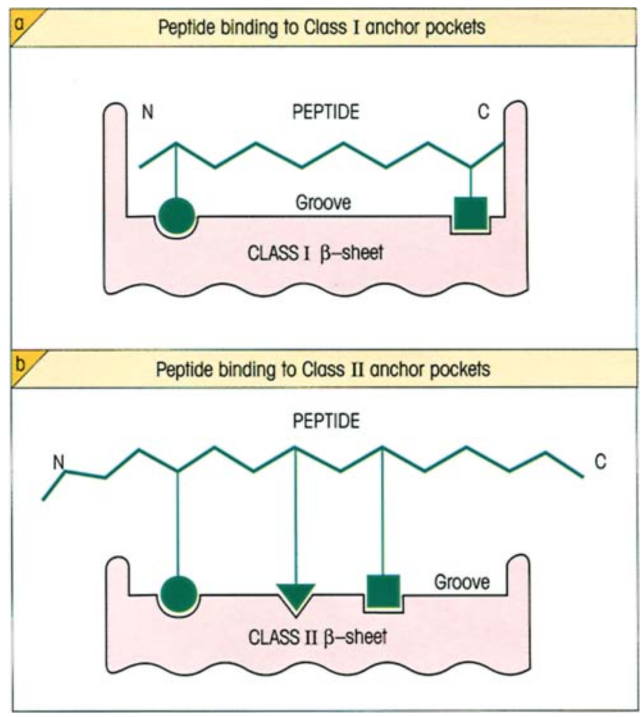

Figure 5.21. Allele-specific pockets in the MHC-binding grooves bind the major anchor residue motifs of the peptide ligands. Cross-section through the longitudinal axis of the MHC groove. The two α-helices forming the lateral walls of the groove lie horizontally above and below the plane of the paper. (a) The class I groove is closed at both ends. The anchor at the carboxy terminus is invariant but the second anchor very often at P2 may also be at P3, P5 or P7 depending on the MHC allele (cf. table 5.2). (b) In contrast, the class II groove is open at both ends and does not constrain the length of the peptide. There are usually three major anchor pockets at P1, P4, P6, P7 or P9 with P1 being the most important.

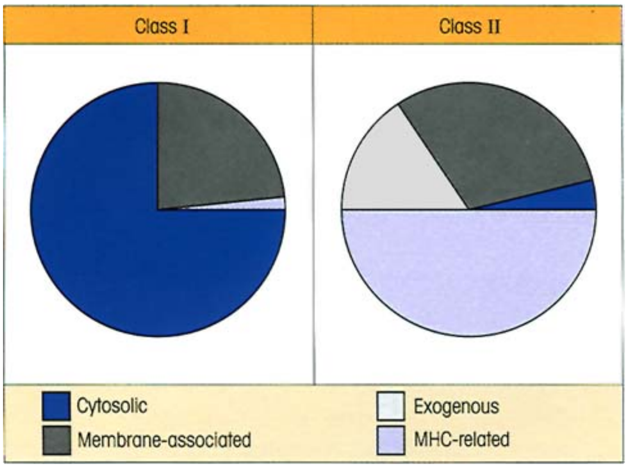

Figure 5.22. The origins of class I- and class II-bound peptides. Virtually all class I peptides are derived from endogenous proteins and, even after viral infections, it is the intracellular antigens which are processed. Processing in the endosomal compartments ensures that proteins of endogenous origin and those derived from membranes constitute over 90% of the peptides bound to the class II grooves. (Diagram reproduced from Vignali D.A.A. & Strominger J.L. (1994) *The Immunologist* **2**, 112, with permission of the authors and publisher.)

ing from eight to 30 amino acid residues, and analysis of such a naturally occurring pool isolated from the MHC would not establish which amino acid side-chains were binding preferentially to the nine available sites within the groove. A modern approach is to study the binding of soluble class II molecules to very large libraries of random-sequence peptides expressed on the surface of bacteriophages (cf. the combinatorial phage libraries, p. 124). The idea is emerging that each amino acid in a peptide contributes independently of the others to the total binding strength, and it should be possible to compute each contribution quantitatively from this random binding data, so that ultimately we could predict which sequences in a given protein antigen would bind to a given class II allele.

Because of the accessible nature of the groove, as the native molecule is unfolded and reduced, but before any degradation need occur, the high affinity epitopes could immediately bury themselves in the class II-binding groove where they are protected from proteolysis. At least for the HLA-DR1 molecule it has been shown that peptide binding leads to a transition from a more open conformation to one with a more compact structure extending throughout the peptide-binding groove. Trimming can take place after peptide binding, leaving peptides 8–30 amino acids long. Several factors will influence the relative concentration of peptide–MHC complex formed: the affinity for the groove as determined by the fit of the anchors, enhancement or hindrance by internal residues (sequences outside the binding residues have little or no effect on peptide-binding specificity), sensitivity to proteases and disulfide reduction, and downstream competition from determinants of higher affinity.

The range of concentration of the different peptide complexes which result will engender a hierarchy of epitopes with respect to their ability to interact with T-cells; the most effective will be **dominant**, the less so **subdominant**. Dominant, and presumably subdominant, **self** epitopes will generally induce tolerance during T-cell ontogeny in the thymus (see p. 231). Complexes with some self peptides which are of relatively low abundance will not tolerize their T-cell counterparts and these autoreactive T-cells constantly pose an underlying threat of potential autoimmunity. Sercarz has labeled these **cryptic** epitopes, and we will discuss their possible relationship to autoimmune disease in Chapter 19.

THE αβ T-CELL RECEPTOR FORMS A TERNARY COMPLEX WITH MHC AND ANTIGENIC PEPTIDE

The forces involved in peptide binding to MHC and in TCR binding to peptide–MHC are similar to those seen between antibody and antigen, i.e. noncovalent. When soluble TCR preparations produced using recombinant DNA technology are immobilized on a sensor chip, they can bind MHC–peptide complex specifically with rather low affinities (K_a) in the 10^4 to $10^7\,\text{M}^{-1}$ range. This low affinity and the relatively small number of atomic contacts (27–68 in the structures so far solved) formed between the TCRs and their MHC–peptide ligands when T-cells contact their target cell make the contribution of TCR recognition to the binding energy of this cellular interaction fairly trivial. The brunt of the attraction rests on the antigen-independent major adhesion molecules, such as ICAM-1/2, LFA-1/2 and CD2, but any subsequent triggering of the T-cell by MHC–peptide antigen must involve signaling through the T-cell receptor.

Topology of the ternary complex

Of the three complementarity determining regions present in each TCR chain, CDR1 and CDR2 are much less variable than CDR3 which, like its immunoglobulin counterpart, has (D)J sequences which result from a multiplicity of combinatorial and nucleotide insertion mechanisms (cf. p. 65). Since the MHC elements in a given individual are fixed, but great variability is expected in the antigenic peptide, a logical model would have CDR1 and CDR2 of each TCR chain contacting the α-helices of the MHC, and the CDR3 concerned in binding to the peptide. In accord with this view, several studies have shown that T-cells which recognize small variations in a peptide in the context of a given MHC restriction element differ only in their CDR3 hypervariable regions.

The combining sites of the TCRs which have been crystallized to date are relatively flat (figure 5.23), which would be expected given the need for complementarity to the gently undulating surface of the peptide–MHC combination. For recognition of peptides presented by MHC class I (figure 5.24a), the TCR lies diagonally across the peptide–MHC with the TCR Vα domain overlying the MHC α_2-helix and the N-terminus of the peptide and the Vβ domain overlying the α_1-helix and the C-terminal portion of the peptide (figure 5.24b). Until many more crystal structures are solved, it is impossible to put forward any hard and fast rules for binding, but it is already clear that the CDR3 loops, which have the greatest variability, make the major contacts with the peptide, particularly focusing in on the middle of the peptide (P4 to P6). The CDR1s can make contacts with both the peptide and the α-helices of the MHC, whilst so far it seems that the CDR2s interact largely with the MHC α-helices (figure 5.24a and b). Significant conformational changes are induced in the CDR3s by these interactions. The recent crystal structure of a TCR recognizing a peptide presented by MHC class II indicates that here there is a different orientation, with the TCR crossing the bound peptide in an orthogonal orientation (figure 5.24c). Whilst the peptides presented by MHC class II molecules are longer, the size of the TCR combining site limits recognition to a maximum of nine sequential peptide residues, and it is again the central residues of the peptide within the MHC groove which are the focus of the TCR's attention. The TCR Vα domain contacts the class II β_1-helix and the Vβ domain the α_1-helix.

One idea is that the TCR 'scans' the MHC and that the binding energy between TCR and the MHC is sufficient to allow the TCR to temporarily dock onto the peptide–MHC complex and interrogate the peptide. If there are sufficient energetically favorable contacts between the TCR and the peptide–MHC, then signaling through the TCR complex can occur. It now seems likely that complexes of two TCR molecules with two MHC–peptide moieties can form and this formation of complexes may be linked to T-cell signaling.

T-CELLS WITH A DIFFERENT OUTLOOK

Nonclassical class I molecules can also present antigen

MHC class I-like molecules

In addition to the highly polymorphic classical MHC class I molecules (HLA-A, B and C in the human and H-2K, D and L in the mouse), there are other loci en-

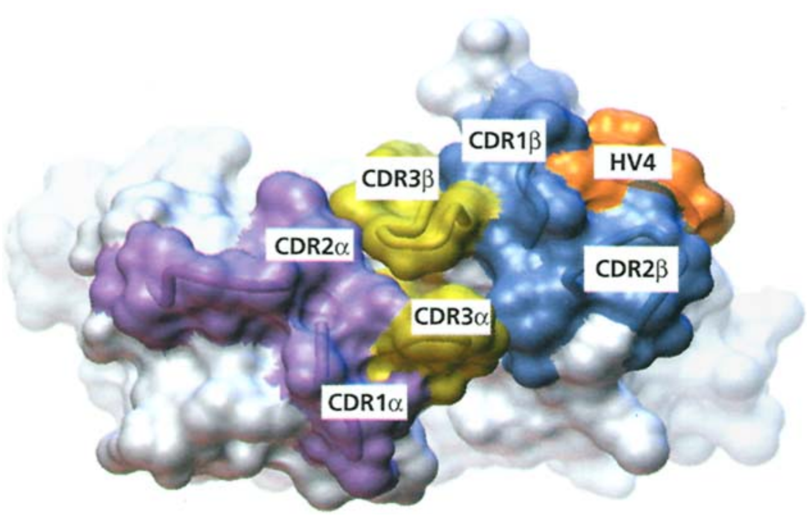

Figure 5.23. T-cell receptor antigen combining site. Although the surface is relatively flat, there is a clearly visible cleft between the CDR3α and CDR3β which can accommodate a central upfacing side-chain of the peptide bound into the groove of an MHC molecule. The surface and loop traces of the Vα CDR1 and CDR2 are colored magenta, Vβ CDR1 and CDR2 blue, Vα CDR3 and Vβ CDR3 yellow, and the Vβ fourth hypervariable region, which makes contact with some superantigens, orange. (Reproduced from Garcia K.C. *et al.* (1996) *Science* **274**, 209, with permission.)

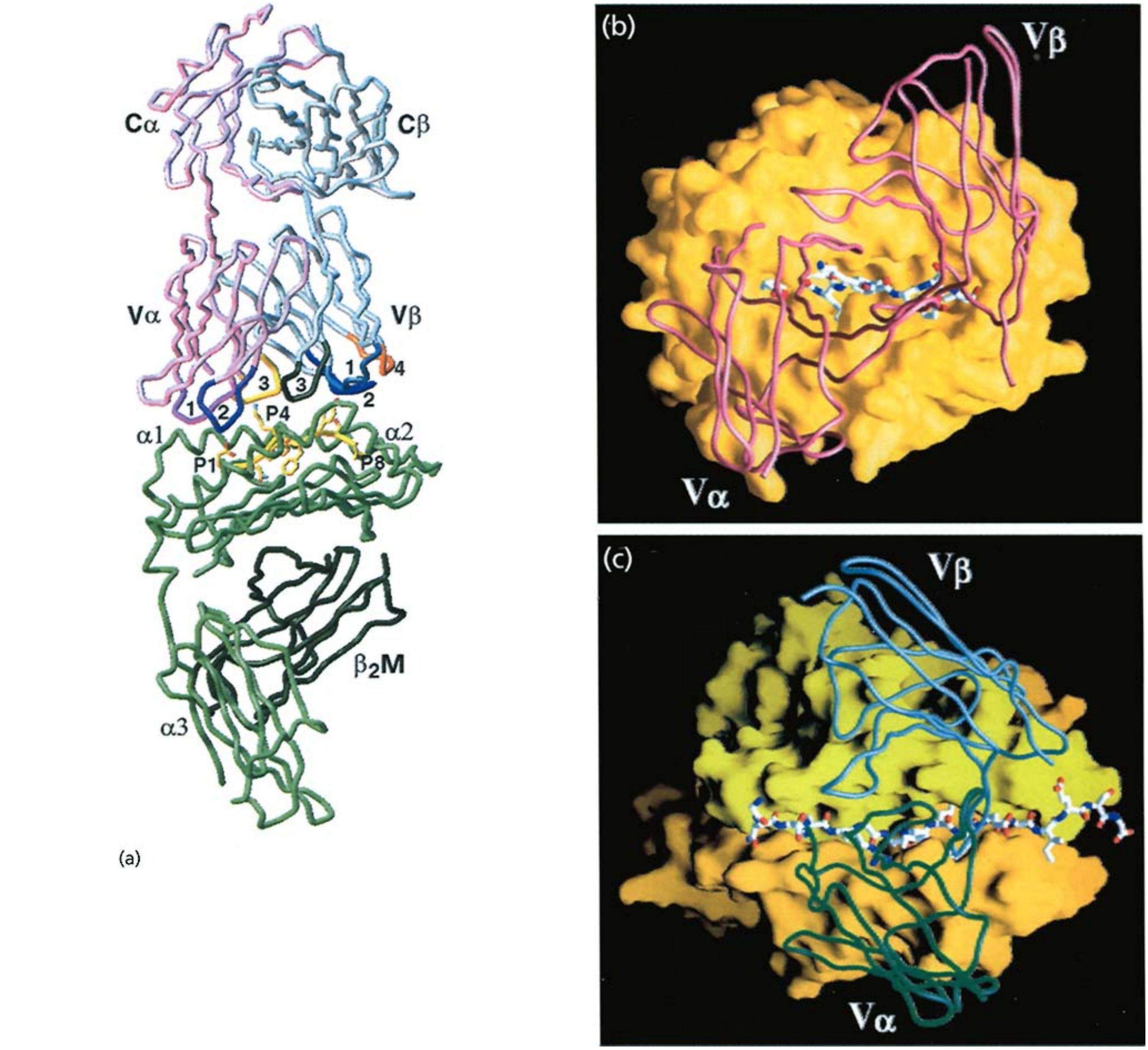

Figure 5.24. Complementarity between MHC–peptide and T-cell receptor. (a) Backbone structure of a TCR (designated 2C) recognizing a peptide (dEV8) presented by the MHC class I molecule H-2K^b. The TCR is in the top half of the picture, with the α chain in pink and its CDR1 colored magenta, CDR2 purple and CDR3 yellow. The β chain is colored light blue with its CDR1 cyan, CDR2 navy blue, CDR3 green and the fourth hypervariable loop orange. Below the TCR is the MHC α chain in green and β_2-microglobulin in dark green. The peptide with its side-chains is colored yellow. (Reproduced from Garcia K.C. *et al.* (1998) *Science* **279**, 1166, with permission.) (b) The same complex looking down onto a molecular surface representation of the H-2K^b in yellow, with the diagonal docking mode of the TCR in a backbone worm representation colored brown. The dEV8 peptide is drawn in a ball and stick format. (c) In contrast, here we see the orthogonal docking mode of a TCR recognizing a peptide presented by MHC class II. The TCR (scD10) backbone worm representation shows the Vα in green and Vβ in blue, and the I-A^k class II molecular surface representation has the α chain in light green and the β chain in orange, holding its conalbumin-derived peptide. (Reproduced from Reinherz E.L. *et al.* (1999) *Science* **286**, 1913, with permission.)

coding MHC molecules containing β_2-microglobulin with relatively nonpolymorphic heavy chains. These are **H-2M**, **Q** and **T** in mice and **HLA-E**, **F** and **G** in *Homo sapiens*.

The best studied molecule encoded by the H-2M locus is **H-2M3**, which is unusual in its ability to present bacterial *N*-formyl methionine peptides to T-cells. Expression of H-2M3 is limited by the availability of these peptides so that high levels are only seen during prokaryotic infections. The demonstration of H-2M3-restricted CD8-positive αβ T-cells specific for *Listeria monocytogenes* encourages the view that this class I-like molecule could underwrite a physiological function in infection. Discussion of the role of HLA-G expression in the human syncytiotrophoblast will arise in Chapter 17 (p. 369).

The family of CD1 non-MHC class I-like molecules can present exotic antigens

After MHC class I and class II, the CD1 family (p. 77) represents a third lineage of antigen-presenting molecules recognized by T-lymphocytes. The CD1 polypeptide chain associates with β_2-microglobulin as becomes an honest class I-like moiety, and the overall structure is similar to that of classical class I molecules, although the topology of the binding groove is altered (see figure 4.18).

CD1 molecules can act as **restriction elements** for the presentation of **lipid and glycolipid** microbial antigens to T-cells. A common structural motif facilitates CD1-mediated antigen presentation and comprises a hydrophobic region of a branched or dual acyl chain and a hydrophilic portion formed by the polar or charged groups of the lipid and/or its associated carbohydrate. The hydrophobic regions are buried in the binding groove of CD1, whilst the hydrophilic regions, such as the carbohydrate structures, are recognized by the TCR. Because antigen recognition by CD1-restricted T-cells involves clonally diverse αβ and γδ TCRs, it is likely that CD1 can present a broad range of such antigens. One group of major ligands for CD1b are glycophosphatidylinositols, such as the mycobacterial cell wall component lipoarabinomannan.

Just like their proteinaceous colleagues, exogenously derived lipid and glycolipid antigens are delivered to the acidic endosomal compartment. Localization of CD1 molecules to the endocytic pathway is mediated by a targeting sequence in the cytoplasmic tail. There is evidence that human CD1a, which lacks the targeting motif, does not localize to this pathway and therefore presumably must follow a different route for peptide loading. The acidic environment of the endosome induces a conformational change in CD1, thereby increasing accessibility to the lipid-binding site in the hydrophobic groove of the molecule. Antigens derived from endogenous pathogens can also be presented by the CD1 pathway, but in a process that, unlike class I-mediated presentation, is independent of TAP and DM.

Some T-cells have NK markers

NK T-cells possess the NK1.1$^+$ marker, characteristic of NK cells, together with a T-cell receptor. However, the TCR bears an invariant α chain (Vα14-Jα281 in mice, Vα24-JαQ in humans) with no N-region modifications and a limited β chain repertoire. These cells are a major component of the T-cell compartment, accounting for 20–30% of T-cells in bone marrow and liver, and up to 1% of spleen cells. NK1.1$^+$ T-cells rapidly secrete interleukin-4 (IL-4) and IFNγ following stimulation and therefore may have important regulatory functions. Although substantial numbers of NK T-cells are CD1-restricted, others are restricted by classical MHC molecules.

γδ TCRs have some features of antibody

Unlike αβ T-cells, γδ T-cells recognize antigens directly without a requirement for antigen processing. Whilst some T-cells bearing a γδ receptor are capable of recognizing MHC molecules, neither the polymorphic residues associated with peptide binding nor the peptide itself are involved. Thus, a γδ T-cell clone specific for the herpes simplex virus glycoprotein-1 can be stimulated by the native protein bound to plastic, suggesting that the cells are triggered by cross-linking of their receptors by antigen which they recognize in the intact native state just as antibodies do. There are structural arguments to give weight to this view. The CDR3 loops, which are critical for foreign antigen recognition by T-cells and antibodies, are comparable in length and relatively constrained with respect to size in the α and β chains of the αβ TCR, presumably reflecting a relative constancy in the size of the MHC–peptide complexes to which they bind. CDR3 regions in the immunoglobulin light chains are short and similarly constrained in length, but in the heavy chains they are longer on average and more variable in length, related perhaps to their need to recognize a wide range of epitopes. Quite strikingly, the γδ TCRs resemble antibodies in that the γ chain CDR3 loops are short with a narrow length distribution, while in the δ chain they are long with a broad length distribution. Therefore, in this respect, the γδ **TCR resembles antibody** more than the αβ TCR. The X-ray crystallographic structure of a TCR Vδ domain highlighted that the γδ TCR indeed incorporates key structural elements of both immunoglobulin and TCR V regions. Overall, the framework regions are more like antibody V_H than TCR $V\alpha$, whilst the conformations and relative positions of the CDRs share features with both $V\alpha$ and V_H domains. The binding site of the $V\delta$ in this particular crystal structure is a relatively flat surface similar to that seen in $\alpha\beta$ TCRs. However, the CDR3 loop of this $V\delta$ is 10 amino acid residues long and, whilst this is approximately the median length of $V\delta$ CDR3s, the broad length distribution already alluded to means that many $\gamma\delta$ receptors will bear longer or shorter $V\delta$ CDR3s. These will have topographically more adventurous binding sites, thereby facilitating the ability of $\gamma\delta$ T-cells to interact with intact rather than processed antigen.

More secrets are being teased out of the $\gamma\delta$ T-cell sect. In the mouse, $\gamma\delta$ T-cells have been isolated which directly recognize the MHC class I molecule I-E^k and the nonclassical MHC molecules T10 and T22 in a peptide-independent fashion. T10 and T22 are expressed by $\alpha\beta$ T-cells following their activation, and it has been suggested that $\gamma\delta$ T-cells specific for these nonclassical MHC molecules may exert a regulatory function. Stressed or damaged cells appear to be powerful activators of $\gamma\delta$ cells, and there is evidence for molecules such as heat-shock proteins as stimulators of $\gamma\delta$ T-cells. Low molecular weight **phosphate-containing nonproteinaceous** antigens, such as isopentenyl pyrophosphate and ethyl phosphate, which occur in a range of microbial and mammalian cells, have been identified as potent stimulators.

A particular subset of $\gamma\delta$ cells, which possess a diverse range of TCRs utilizing different *D* and *J* gene segments but always using the same *V* gene segments, *Vγ2* and *Vδ2*, expand *in vivo* to comprise a large proportion (8–60%) of all peripheral blood T-cells during a diverse range of infections. These $V\gamma2V\delta2$ T-cells recognize a newly appreciated group of antigens, the alkylamines, which have chemical and biological properties distinct from lipid and phosphate antigens, further extending the variety of nonprotein antigens which can be recognized by T-cells. A number of alkylamine antigens are produced by human pathogens, including *Salmonella typhimurium*, *Listeria monocytogenes*, *Yersinia enterocolitica* and *Escherichia coli*. Individual $V\gamma2V\delta2$ T-cells can recognize both positively charged alkylamines and negatively charged molecules such as ethyl phosphate, but this should be fairly straightforward for the receptor given the small hapten-like size of these antigens.

The above characteristics provide the $\gamma\delta$ cells with a distinctive role complementary to that of the $\alpha\beta$ population and enable them to function in the recognition of microbial pathogens and of damaged or stressed host cells.

SUPERANTIGENS STIMULATE WHOLE FAMILIES OF LYMPHOCYTE RECEPTORS

Bacterial toxins represent one major group of T-cell superantigens

Whereas an individual peptide complexed to MHC will react with antigen-specific T-cells which represent a relatively small percentage of the T-cell pool because of the requirement for specific binding to particular CDR3 regions, a special class of molecule has been identified which stimulates the 5–20% of the total T-cell population expressing the same TCR $V\beta$ family structure. These molecules do this irrespective of the antigen specificity of the receptor. They have been described as **superantigens** by Kappler and Marrack.

The pyogenic toxin superantigen family can cause food poisoning, vomiting and diarrhea and includes *Staphylococcus aureus* enterotoxins (SEA, SEB and several others), staphylococcal toxic shock syndrome toxin-1 (TSST-1), streptococcal superantigen (SSA) and several streptococcal pyogenic exotoxins (SPEs). Although these molecules all have a similar structure, they stimulate T-cells bearing different $V\beta$ sequences. They are strongly mitogenic for these T-cells in the presence of MHC class II accessory cells. SEA must be one of the most potent T-cell mitogens known, causing marked proliferation in the concentration range 10^{-13} to 10^{-16} M. Like the other superantigens it can cause the release of copious amounts of cytokines, including IL-2 and lymphotoxin, and of mast cell leukotrienes, which probably form the basis for its ability to produce toxic shock syndrome. Other superantigens which do not belong to the pyogenic toxin superantigen family include staphylococcal exfoliative toxins (ETs), *Mycoplasma arthritidis* mitogen (MAM) and *Yersinia pseudotuberculosis* mitogen.

Superantigens are not processed by the antigen-presenting cell, but cross-link the class II and $V\beta$ independently of direct interaction between MHC and TCR molecules (figure 5.25).

Endogenous mouse mammary tumor viruses (MMTV) act as superantigens

Very many years ago, Festenstein made the curious observation that B-cells from certain mouse strains could produce powerful proliferative responses in roughly 20% of unprimed T-cells from another strain of identical MHC. The so-called Mls gene product responsible for inciting proliferation turns out to be encoded by the open reading frame (ORF) located in the 3′ long terminal repeat of **MMTV**. They are type B retroviruses transmitted as infectious agents in milk and are specific for B-cells. They associate with class II MHC in the B-cell membrane and act as superantigens through their affinity for certain TCR Vβ families in a similar fashion to the bacterial toxins. Other proposed viral superantigens capable of polyclonally activating T-cells include the nucleocapsid protein of rabies virus, and antigens associated with cytomegalovirus and with Epstein–Barr virus.

Microbes can also provide B-cell superantigens

Staphylococcal protein A reacts not only with the Fcγ region of IgG but also with 15–50% of polyclonal IgM, IgA and IgG $F(ab')_2$, all of which belong to the V_H3 family. This superantigen is mitogenic for B-cells through its recognition by a discontinuous binding sequence composed of amino acid residues from FR1, CDR2 and FR3 of the V_H domain. The human immunodeficiency virus (HIV) glycoprotein gp120 also reacts with immunoglobulins which utilize V_H3 family members. The binding site appears to partially overlap with that for protein A and utilizes amino acid residues from FR1, CDR1, CDR2 and FR3.

THE RECOGNITION OF DIFFERENT FORMS OF ANTIGEN BY B- AND T-CELLS IS ADVANTAGEOUS TO THE HOST

It is our conviction that this section deals with a subject of the utmost importance, which is at the epicenter of immunology.

Antibodies combat microbes and their products in the extracellular body fluids where they exist essen-

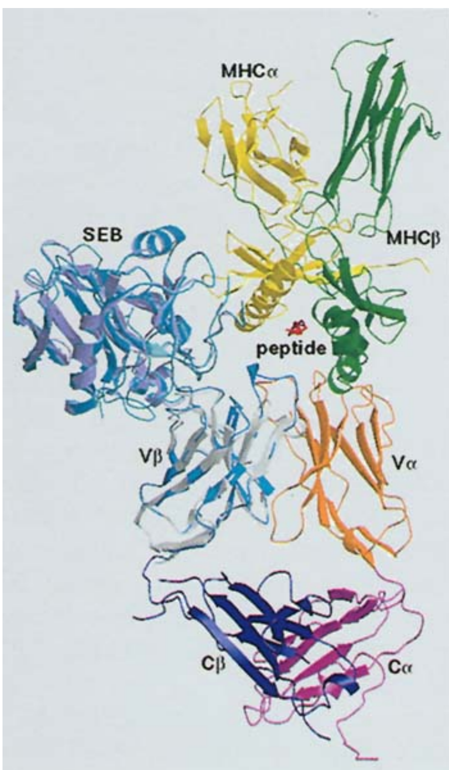

Figure 5.25. Interaction of superantigen with MHC and TCR. In this composite model, the interaction with the superantigen staphylococcal enterotoxin B (SEB) involves SEB wedging itself between the TCR Vβ chain and the MHC, effectively preventing interaction between the TCR and the peptide in the groove, and between the TCR β chain and the MHC. Thus direct contact between the TCR and the MHC is limited to Vα amino acid residues. (Reproduced from Li H. *et al.* (1999) *Annual Review of Immunology* **17**, 435, with permission.) Other superantigens are likely to disrupt direct TCR interactions with peptide–MHC to varying extents.

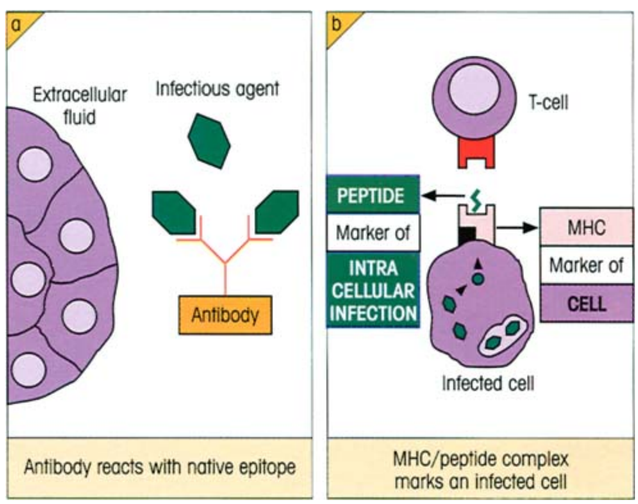

Figure 5.26. (a) Antibodies are formed against the native, not denatured, form of infectious agents which are attacked in the extracellular fluids. (b) Effector T-cells recognize infected cells by two surface markers: the MHC is a signal for the cell, and the foreign peptide is present in the MHC groove since it is derived from the proteins of an intracellular infectious agent. Further microbial cell surface signals can be provided by undegraded antigens and low molecular weight phosphate-containing antigens (seen by γδ T-cells), and lipids and glycolipids presented by CD1 molecules.

tially in their native form (figure 5.26a). Clearly it is to the host's advantage for the B-cell receptor to recognize epitopes on the **native molecules**.

αβ T-cells have quite a different job. In the case of cytotoxic T-cells, and the T-cells which secrete cytokines that activate infected macrophages, they have to seek out and bind to the infected cells and carry out their effector function face to face with the target. First, with respect to proteins produced by intracellular infectious agents, the MHC molecules tell the effector T-lymphocyte that it is encountering a cell. Second, the T-cell does not want to attack an uninfected cell on whose surface a native microbial molecule is sitting adventitiously nor would it wish to have its antigenic target on the appropriate cell surface blocked by an excess of circulating antibody. Thus it is of benefit for the infected cell to express the microbial antigen on its surface in a form distinct from that of the native molecule. As will now be more than abundantly clear, the evolutionary solution was to make the T-cell recognize a processed peptide derived from the intracellular antigen and to hold it as a complex with the surface MHC molecules. The single T-cell receptor then recognizes both the **MHC cell marker** and the **peptide infection marker** in one operation (figure 5.26b).

A comparable situation arises when CD1 molecules substitute for MHC in antigen presentation to T-cells, in this case associating with processed microbial lipids and glycolipids. The physiological role of the γδ cells has yet to be fully unraveled.

SUMMARY

The nature of antigen recognition by antibody

- An antigen is defined by its antibody. The contact area with an antibody is called an **epitope** and the corresponding area on an antibody, a **paratope**.
- Antisera recognize a series of dominant epitope clusters on the surface of an antigen; each cluster is called a determinant.
- Most epitopes on globular proteins are **discontinuous** rather than **linear**, involving amino acids far apart in the primary sequence.
- The protruding regions and probably the 'flexible' segments of globular proteins tend to be associated with higher epitope densities.

Antigens and antibodies interact by spatial complementarity, not by covalent binding

- The forces of interaction include electrostatic, hydrogen bonding, hydrophobic and van der Waals.
- The forces become large as the separation of antigen and antibody diminishes, especially when water molecules are excluded.
- Antigen–antibody bonds are readily **reversible**.
- Antigens and antibodies are mutually deformable.

Affinity

- The strength of binding to a single antibody combining site is measured by the **affinity**.
- The reaction of multivalent antigens with the heterogeneous mixture of antibodies in an antiserum is defined by **avidity (functional affinity)** and is usually much greater than affinity due to the 'bonus effect of multivalency'.
- The **specificity** of antibodies is not absolute and they may cross-react with other antigens to different extents, measured by their relative avidities.

T-cell recognition

- αβ T-cells see antigen in association with MHC molecules.
- They are restricted to the haplotype of the cell which first primed the T-cell.
- Protein antigens are processed by antigen-presenting cells to form small linear peptides which associate with the MHC molecules, binding to the central groove formed by the α-helices and the β-sheet floor.

Processing of antigen for presentation by class I MHC

- Endogenous cytosolic antigens such as viral proteins are cleaved by **immunoproteasomes** and the peptides so formed are **transported** to the ER by the TAP1/2 system.
- The peptide then dissociates from TAP1/2 and forms a stable heterotrimer with newly synthesized class I MHC heavy chain and β_2-microglobulin.
- This **peptide–MHC complex** is then transported to the surface for presentation to cytotoxic T-cells.

(*continued p. 106*)

Processing of antigen for presentation by class II MHC

- The αβ **class II molecule** is synthesized in the ER and complexes with membrane-bound **invariant chain (Ii)**.
- This facilitates transport of the vesicles containing class II across the Golgi and directs them to an acidified late endosome or lysosome containing exogenous protein taken into the cell by endocytosis or phagocytosis.
- Proteolytic degradation of Ii in the endosome leaves a peptide referred to as CLIP which protects the MHC groove.
- Processing by endosomal proteases degrades the antigen to peptides which replace the CLIP.
- The **class II–peptide** complex now appears on the cell surface for presentation to T-helper cells.

The nature of the peptide

- Class I peptides are held in extended conformation within the MHC groove.
- They are usually 8–9 residues in length and have two or three key **anchor**, relatively invariant residues which bind to allele-specific pockets in the MHC.
- Class II peptides are between 8 and 30 residues long, extend beyond the groove and usually have three or four anchor residues.
- The other amino acid residues in the peptide are greatly variable and are recognized by the T-cell receptor (TCR).

Complex between TCR, MHC and peptide

- The first and second hypervariable regions (CDR1 and CDR2) of each TCR chain mostly contact the MHC α-helices, while the CDR3s, having the greatest variability, interact with the antigenic peptide. Complexes of TCR_2 with $(MHC–peptide)_2$ are probably formed.

Some T-cells are independent of classical MHC molecules

- MHC class I-like molecules, such as H-2M, are relatively nonpolymorphic and can present antigens such as bacterial *N*-formyl methionine peptides.
- The CD1 family of non-MHC class I-like molecules can present antigens such as lipid and glycolipid mycobacterial antigens.
- γδ T-cells resemble antibodies in recognizing whole unprocessed molecules such as low molecular weight phosphate-containing nonproteinaceous molecules.

Superantigens

- These are potent mitogens which stimulate whole lymphocyte subpopulations sharing the same TCR Vβ or immunoglobulin V_H family independently of antigen specificity.
- *Staphylococcus aureus* enterotoxins are powerful human superantigens which cause food poisoning and toxic shock syndrome.
- T-cell superantigens are not processed but cross-link MHC class II and TCR Vβ independently of their direct interaction.
- Mouse mammary tumor viruses are B-cell retroviruses which are superantigens in the mouse.

Recognition of different forms of antigen by B- and T-cells is an advantage

- B-cells recognize epitopes on the native antigen; this is important because antibodies react with native antigen in the extracellular fluid.
- T-cells must contact infected cells and, to avoid confusion between the two systems, the infected cell signals itself to the T-cell by the combination of MHC and degraded antigen.

See the accompanying website (**www.roitt.com**) for multiple choice questions.

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Immunochemical techniques

INTRODUCTION

Since antigens and antibodies are defined by their mutual interactions, they can each be used to quantify each other. Before we get down to details, it is worth posing the question 'What does serum "antibody content" mean?'

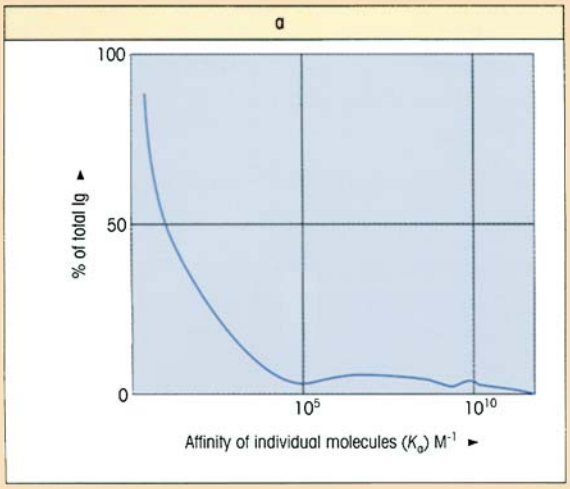
If we have a solution of a monoclonal antibody, we can define its affinity and specificity with considerable confidence and, if pure and in its native conformation, we will know that the concentration of antibody is the same as that of the measurable immunoglobulin in ng/ml or whatever. When it comes to measuring the antibody content of an antiserum, the problem is of a different order because the immunoglobulin fraction is composed of an enormous array of molecules of varying abundance and affinity (figure 6.1a).

An **average** K_a for the whole IgG can be obtained by analysing the overall interaction with antigen as a mass action equation. But how can the **antibody content** of the IgG be defined in a meaningful way? The answer is of course that one would usually wish to describe antibody in practical functional terms: does a serum protect against a given infectious dose of virus, does it promote effective phagocytosis of bacteria, does it permit complement-mediated bacteriolysis, does it neutralize toxins, and so on? For such purposes, very low affinity molecules would be useless because they form such inadequate amounts of complex with the antigen.

At the practical level in a diagnostic laboratory, the functional tests are labor intensive and therefore expensive, and a compromise is usually sought by using immunochemical assays which measure a composite of medium to high affinity antibodies and their abundance. The majority of such tests usually measure the total amount of antibody binding to a given amount of antigen; this could be a modest amount of high affinity antibody or much more antibody of lower affinity, or all combinations in between. Sera are compared for high or low 'antibody content' either by seeing how much antibody binds to antigen at a fixed serum dilution, or testing a series of serum dilutions to see at which level a standard amount of antibody just sufficient to give a positive result is bound. This is the so-called **antibody titer**. To take an example, a serum might be diluted, say, 10 000 times and still just give a positive agglutination test (cf. figure 6.9). This titer of 1 : 10 000 enables comparison to be made with another much 'weaker' serum which has a titer of only, say, 1 : 100. Note that the titer of a given serum will vary with the sensitivity of the test, since much smaller amounts of antibody are needed to bind to antigen for a highly sensitive test, such as agglutination, than for a test of low sensitivity, such as precipitation, which requires high concentrations of antibody–antigen product (figure 6.1b).

To summarize: the '**effective antibody contents**' of different sera can be compared by seeing **how**

(*continued*)



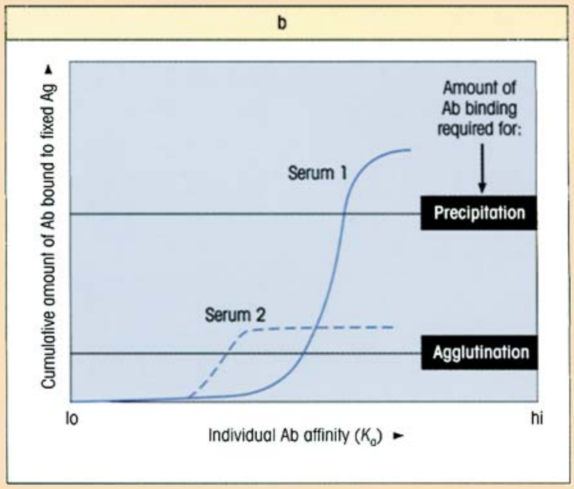


Figure 6.1. Distribution of affinity and abundance of IgG molecules in an individual serum. (a) Distribution of affinities of IgG molecules for a given antigen in the serum of a hypothetical individual. There is a great deal of low affinity antibody which would be incapable of binding to antigen effectively, and much lower amounts of high affinity antibody whose skewed distribution is assumed to arise from exposure to infection. (b) Relationship of affinity distribution to positivity in tests for antigen binding. Rearranging the mass action equation, for all molecules of the same affinity K_x and concentration of unbound antibody $[Ab_x]$:

the amount of complex formed $[AgAb] \propto K_x[Ab_x]$ for fixed $[Ag]$.

Starting with the lowest affinity molecules in the serum, we have charted the cumulative total of antibody bound for each antibody species up to and including the one being plotted. As might be expected, the very low affinity antibodies make no contribution to the tests. Serum 2 has more low affinity antibody and virtually no high affinity, but it can produce just enough complex to react in the sensitive agglutination test although, unlike serum 1, it forms insufficient to give a positive precipitin. Because of its relatively high 'content' of antibody, serum 1 can be diluted to a much greater extent than serum 2 and yet still give positive agglutination, i.e. it has a higher titer. The precipitin test is less sensitive, requiring more complex formation, and serum 1 cannot be diluted much before this test becomes negative, i.e. the precipitin titer will be far lower than the agglutination titer for the same serum.

much antibody binds to the fixed amount of test antigen, or the titer can be determined, i.e. **how far the serum can be diluted before the test becomes negative.** This is a compromise between abundance and affinity and for practical purposes is used as an approximate indicator of biological effectiveness.

ESTIMATION OF ANTIBODY

Antigen–antibody interactions in solution

The classical precipitin reaction

When an antigen solution is added progressively to a potent antiserum, antigen–antibody precipitates are formed (figure 6.2a and b). The cross-linking of antigen and antibody gives rise to three-dimensional lattice structures, as suggested by John Marrack, which coalesce, largely through Fc–Fc interaction, to form large precipitating aggregates. As more and more antigen is added, an optimum is reached (figure 6.2b) after which consistently less precipitate is formed. At this stage the supernatant can be shown to contain soluble complexes of antigen (Ag) and antibody (Ab), many of composition Ag_4Ab_3, Ag_3Ab_2 and Ag_2Ab (figure 6.2c). In extreme antigen excess (figure 6.2c), ultracentrifugal analysis reveals the complexes to be mainly of the form Ag_2Ab, a result directly attributable to the two combining sites (divalence) of the IgG antibody molecule (cf. electron microscope study, figure 3.1, and Scatchard analysis, figure 5.10a).

Serums frequently contain up to 10% of nonprecipitating antibodies which are effectively monovalent because of the asymmetric presence of oligosaccharide on one antigen-binding arm of the antibody molecule which stereochemically blocks the combining site. Also, frank precipitates are only observed when antigen, and particularly antibody, are present in fairly hefty concentrations. Thus, when complexes are formed which do not precipitate spontaneously, more devious methods must be applied to detect and estimate the antibody level.

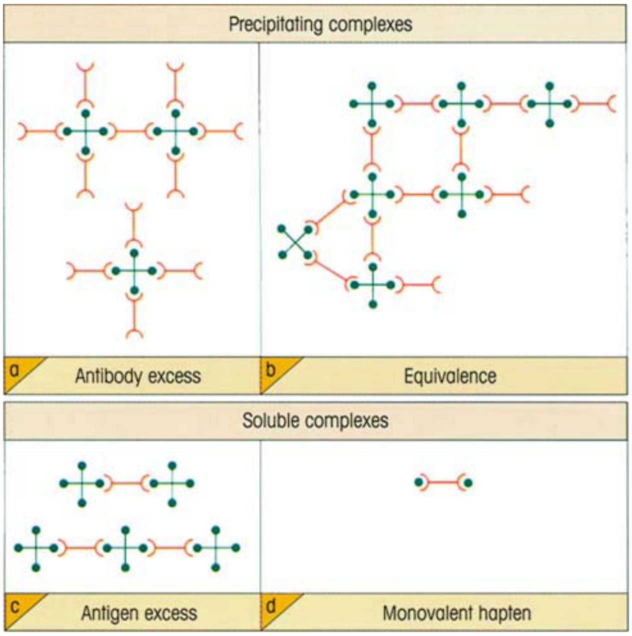


Figure 6.2. Diagrammatic representation of complexes formed between a hypothetical tetravalent antigen ( ) and bivalent antibody ( ) mixed in different proportions. In practice, the antigen valencies are unlikely to lie in the same plane or to be formed by identical determinants as suggested in the figure. (a) In extreme antibody excess, the antigen valencies are saturated and the molar ratio Ab:Ag approximates to the valency of the antigen. (b) At equivalence, most of the antigen and antibody combines to form large lattices which aggregate to produce typical immune precipitates. (c) In extreme antigen excess, where the two valencies of each antibody molecule become rapidly saturated, the complex Ag_2Ab tends to predominate. (d) A monovalent hapten binds but is unable to cross-link antibody molecules.

Nonprecipitating antibodies can be detected by nephelometry

The small aggregates formed when dilute solutions of antigen and antibody are mixed create a cloudiness or turbidity which can be measured by forward angle scattering of an incident light source (nephelometry). Greater sensitivity can be obtained by using monochromatic light from a laser and by adding polyethylene glycol to the solution so that aggregate size is increased. In practice, nephelometry is applied more to the detection of antigen than antibody and this will be dealt with in a later section.

Complexes formed by nonprecipitating antibodies can be precipitated

The relative antigen-binding capacity of an antiserum which forms soluble complexes can be estimated using radiolabeled antigen. The complex can be brought out of solution either by changing its solubility or by adding an anti-immunoglobulin reagent as in figure 6.3.

Enhancement of precipitation by countercurrent immunoelectrophoresis

This technique may be applied to antigens which migrate towards the positive pole on electrophoresis in agar (if necessary antigens can be substituted with negatively charged groups to achieve this end). Antigen and antiserum are placed in wells punched in the agar gel and a current applied (figure 6.4). The antigen migrates steadily into the antibody zone where it successively binds more and more antibody molecules, in essence artificially increasing the effective antibody concentration and thus forming a precipitin line.

Measurement of antibody affinity

As discussed in earlier chapters (cf. p. 85), the binding strength of antibody for antigen is measured in terms of the association constant (K_a) or its reciprocal, the dissociation constant (K_d), governing the reversible interaction between them and defined by the mass action equation at equilibrium:

$$K_a = \frac{[\text{AgAb complex}]}{[\text{free Ag}][\text{free Ab}]}$$

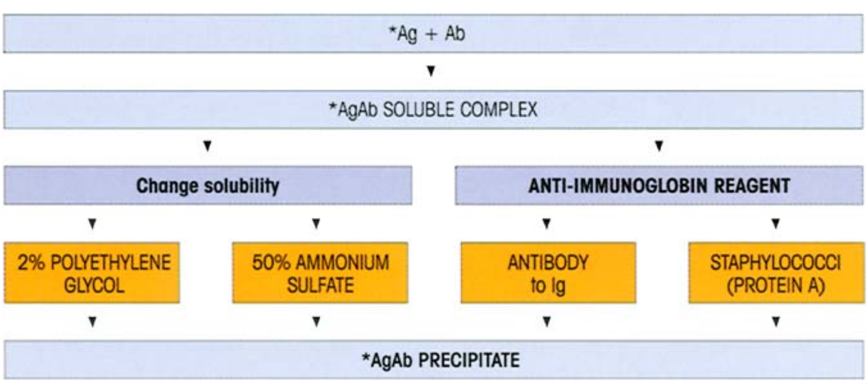


Figure 6.3. Binding capacity of an antiserum for labeled antigen (*Ag) by precipitation of soluble complexes either: (i) by changing the solubility so that the complexes are precipitated while the uncombined Ag and Ab remain in solution, or (ii) by adding a precipitating anti-immunoglobulin antibody or staphylococcal organisms which bind immunoglobulin Fc to the protein A on their surface; the complex can then be spun down. The level of label (e.g. radioactivity) in the precipitate will be a measure of antigen-binding capacity.

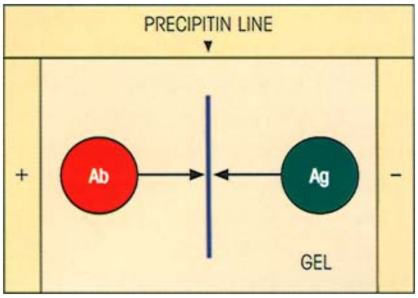


Figure 6.4. Countercurrent immunoelectrophoresis. Antibody moves towards the negative pole in the gel on electrophoresis due to endosmosis; an antigen which is negatively charged at the pH employed will move towards the positive pole and precipitate on contact with antibody.

With small haptens, the equilibrium dialysis method can be employed to measure K_a (see p. 87), but usually one is dealing with larger antigens and other techniques must be used. One approach is to add increasing amounts of radiolabeled antigen to a fixed amount of antibody, and then separate the free from bound antibody by precipitating the soluble complex as described above (e.g. by an anti-immunoglobulin). The reciprocal of the bound, i.e. complexed, antibody concentration can be plotted against the reciprocal of the free antigen concentration, so allowing the affinity constant to be calculated (figure 6.5a). For an antiserum this will give an affinity constant representing

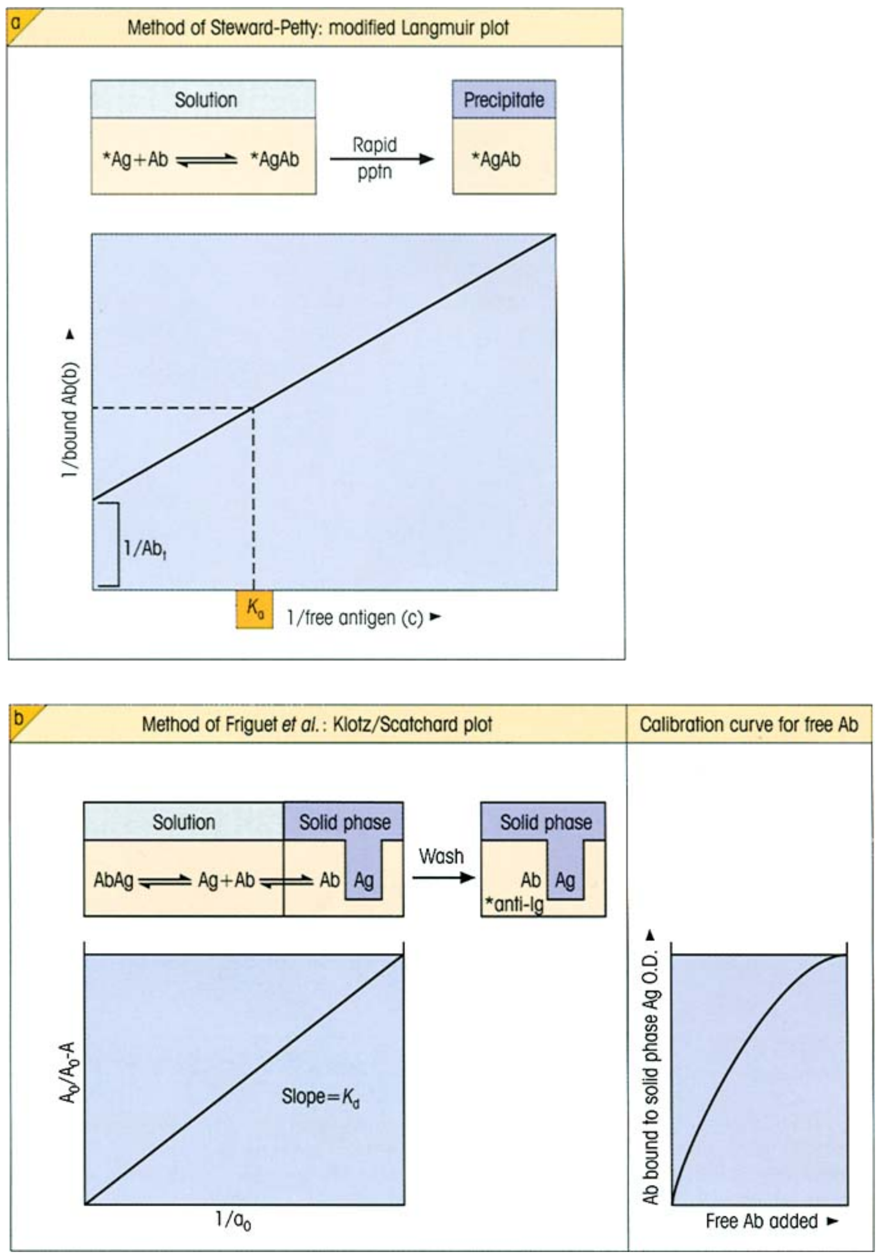


Figure 6.5. Determination of affinity with large antigens. The equilibria between Ab and Ag at different concentrations are determined as follows:
(a) For a polyclonal antiserum one can use the Steward–Petty modification of the Langmuir equation:

$1/b = 1/(Ab_t.c.K_a) + 1/Ab_t$

where Ab_t = total Ab combining sites, b = bound Ab concentration, c = free Ag concentration and K_a = average affinity constant. At infinite Ag concentration, all Ab sites are bound and $1/b = 1/Ab_t$. When half the Ab sites are bound, $1/c = K_a$ (cf. p. 86).
(b) The method of Friguet *et al.* for monoclonal antibodies. First, a calibration curve for free antibody is established by estimating the proportion binding to solid-phase antigen, bound antibody being measured by enzyme-labeled anti-Ig (ELISA: see text). Using the calibration curve, the amount of free Ab in equilibrium with Ag in solution is determined by seeing how much of the Ab binds to solid-phase Ag (the amount of solid-phase antigen is insufficient to affect the solution equilibrium materially). Combination of the Klotz and Scatchard equations gives:

$A_o/A_o - A = 1 + K_d/a_o$

where A_o = ELISA optical density (OD) for Ab in the absence of Ag, A = OD in the presence of Ag concentration a_o where a_o is approximately 10 × concentration of Ab. The slope of the plot gives K_d. (Labeled molecules are marked with an asterisk.)

an average of the heterogeneous antibody components and a measure of the effective number of antigen-binding sites operative at the highest levels of antigen used.

Various types of ELISA (see below) have been developed which provide a measure of antibody affinity. In one system the antibody is allowed to first bind to its antigen, and then a chaotropic agent such as thiocyanate is added in increasing concentration in order to disrupt the antibody binding; the higher the affinity of the antibody, the more agent that is required to reduce the binding. Another type of ELISA for measuring affinity is the indirect competitive system devised by Friguet and associates (figure 6.5b). A constant amount of antibody is incubated with a series of antigen concentrations and the free antibody at equilibrium is assessed by secondary binding to solid-phase antigen. In this way, values for K_a are not affected by any distortion of antigen by labeling. This again stresses the superiority of determining affinity by studying the **primary reaction** with antigen in the **soluble state** rather than conformationally altered through binding to a solid phase.

Increasingly, affinity measurements are obtained using **surface plasmon resonance**. A sensor chip consisting of a monoclonal antibody coupled to dextran overlying a gold film on a glass prism will totally internally reflect light at a given angle (figure 6.6a). Antigen present in a pulse of fluid will bind to the sensor chip and, by increasing its size, alter the angle of reflection. The system provides data on the kinetics of association and dissociation (and hence K) (figure 6.6b) and permits comparisons between monoclonal antibodies and also assessment of subtle effects of mutations.

Agglutination of antigen-coated particles

Whereas the cross-linking of multivalent protein antigens by antibody leads to precipitation, cross-linking of cells or large particles by antibody directed against surface antigens leads to agglutination. Since most cells are electrically charged, a reasonable number of antibody links between two cells are required before the mutual repulsion is overcome. Thus agglutination of cells bearing only a small number of determinants may be difficult to achieve unless special methods such as further treatment with an antiglobulin reagent are used. Similarly, the higher avidity of multivalent IgM antibody relative to IgG (cf. p. 89) makes the former more effective as an agglutinating agent, molecule for molecule (figure 6.7).

Agglutination reactions are used to identify bacteria and to type red cells; they have been observed with

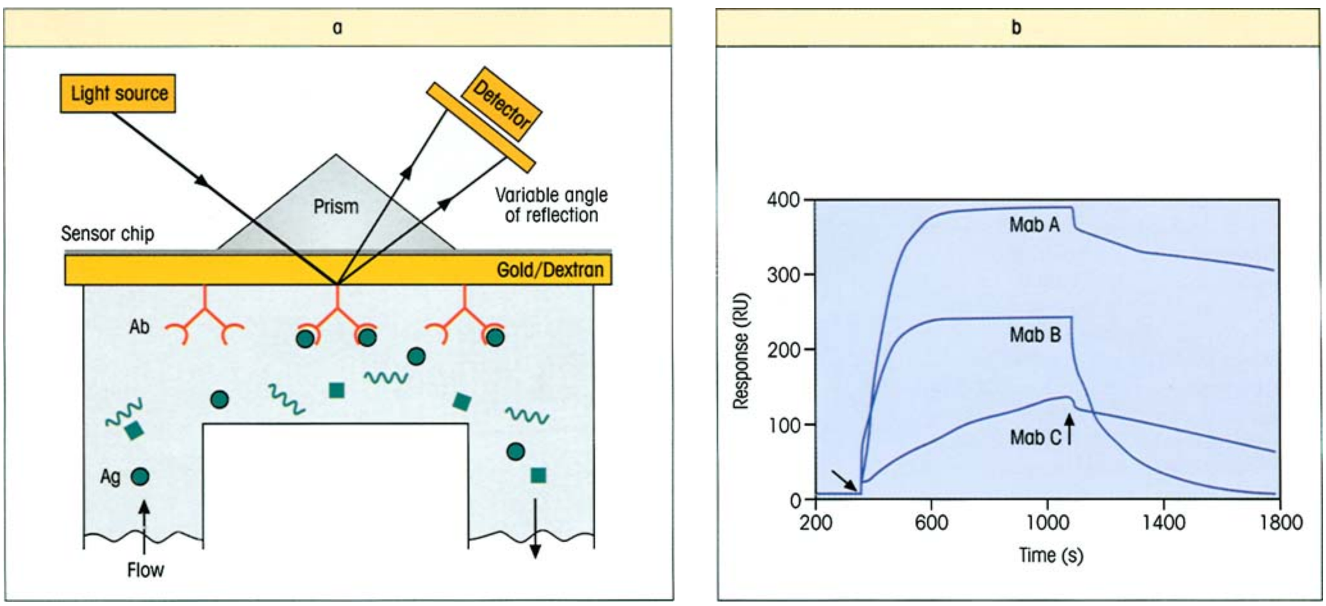


Figure 6.6. Surface plasmon resonance. (a) The principle: as antigen binds to the antibody-coated sensor chip it alters the angle of reflection. (b) This signals the rates of association during the antigen pulse and dissociation. In this example, the same antigen was injected over three immobilized monoclonal antibodies. The arrows point to the beginning and end of the antigen injection, which is followed by buffer flow. Note the differences between the antibodies in the association and dissociation rates. (Data kindly provided by Dr R. Karlsson, Biacore AB, and reproduced from Panayotou G. (1998) Surface plasmon resonance. In Delves P.J. & Roitt I.M. (eds) *Encyclopedia of Immunology*, 2nd edn. Academic Press, with permission.) The system can be used with antigen immobilized on the sensor chip and antibody in the fluid phase, or can be applied to any other single ligand-binding assay.

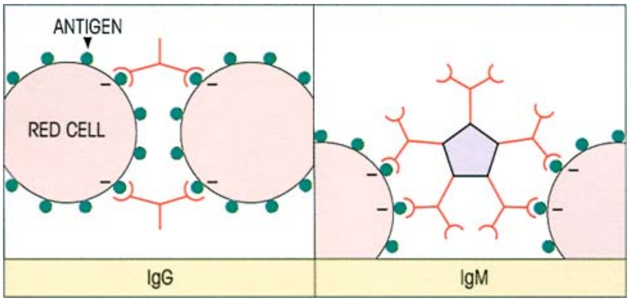
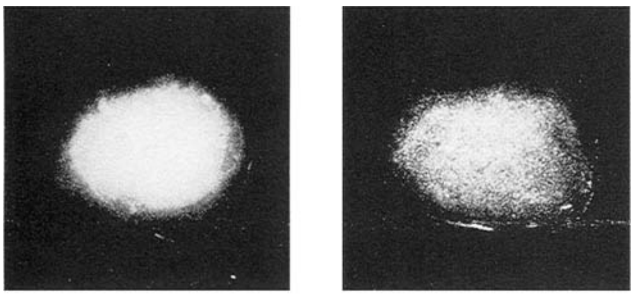


Figure 6.7. Mechanism of agglutination of antigen-coated particles by antibody cross-linking to form large macroscopic aggregates. If red cells are used, several cross-links are needed to overcome the electrical charge at the cell surface. IgM is superior to IgG as an agglutinator because of its multivalent binding and because the charged cells are further apart.



(a) (b)

Figure 6.8. Macroscopic agglutination of latex coated with human IgG by serum from a patient with rheumatoid arthritis. This contains rheumatoid factor, an autoantibody directed against determinants on IgG. (a) Normal serum. (b) Patient's serum.

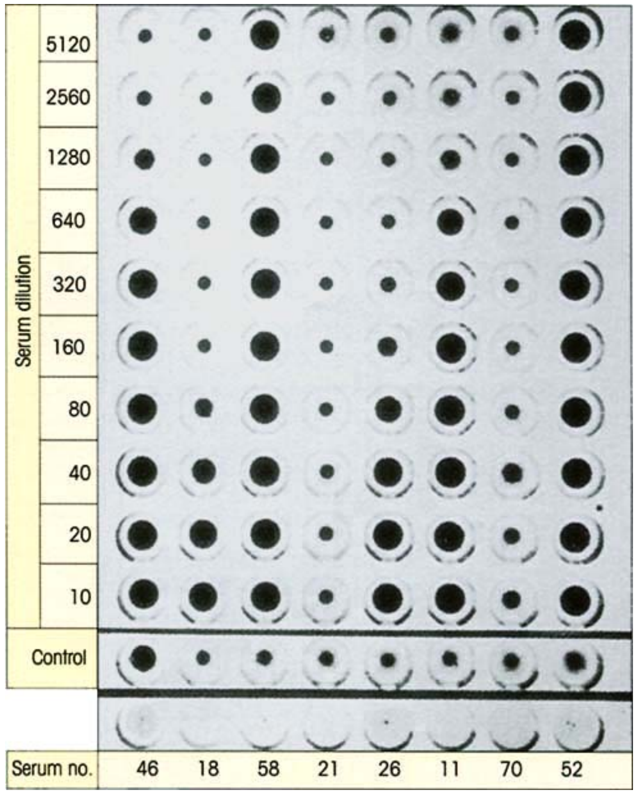


Figure 6.9. Red cell hemagglutination test for thyroglobulin autoantibodies. Thyroglobulin-coated cells were added to dilutions of patients' serums. Uncoated cells were added to a 1 : 10 dilution of serum as a control. In a positive reaction, the cells settle as a carpet over the bottom of the cup. Because of the 'V'-shaped cross-section of these cups, in negative reactions the cells fall into the base of the 'V', forming a small, easily recognizable button. The reciprocal of the highest serum dilution giving an unequivocally positive reaction is termed the titer. The titers reading from left to right are: 640, 20, >5120, neg, 40, 320, neg, >5120. The control for serum no. 46 was slightly positive and this serum should be tested again after absorption with uncoated cells.

leukocytes and platelets, and even with spermatozoa in certain cases of male infertility due to sperm agglutinins. Because of its sensitivity and convenience, the test has been extended to the identification of antibodies to soluble antigens which have been artificially coated on to erythrocytes, latex or gelatin particles. Agglutination of IgG-coated latex is used to detect rheumatoid factors (figure 6.8). Similar tests using antigen-coated particles can be carried out in U-bottom microtiter plates where the settling pattern on the bottom of the well may be observed (figure 6.9); this provides a more sensitive indicator than macroscopic clumping. Quantification of more subtle degrees of agglutination can be achieved by nephelometry or Coulter counting.

Immunoassay for antibody using solid-phase antigen

The principle

The antibody content of a serum can be assessed by the ability to bind to antigen which has been immobilized by physical adsorption to a plastic tube or microtiter plate with multiple wells; the bound immunoglobulin may then be estimated by addition of a labeled anti-Ig raised in another species (figure 6.10). Consider, for example, the determination of DNA autoantibodies in SLE (cf. p. 401). When a patient's serum is added to a microwell coated with antigen (in this case DNA), the autoantibodies will bind to the antigen and the remaining serum proteins can be readily washed away. Bound antibody can now be estimated by addition of ^{125}I-labeled purified rabbit anti-human IgG; after rinsing out excess unbound reagent, the radioactivity of the tube will clearly be a measure of the autoantibody content of the patient's serum. The distribution of antibody in different classes can be determined by using specific antisera. Take the radioallergosorbent test (RAST) for IgE antibodies in allergic patients. The allergen (e.g. pollen extract) is covalently coupled to an immunoabsorbent, in this case a

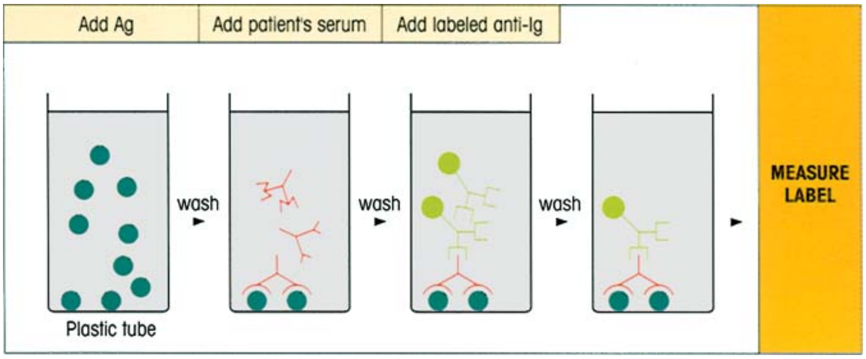


Figure 6.10. Solid-phase immunoassay for antibody. To reduce nonspecific binding of IgG to the solid phase after adsorption of the first reagent, it is usual to add an irrelevant protein, such as dried skimmed milk powder or bovine serum albumin, to block any free sites on the plastic. Note that the conformation of a protein often alters on binding to plastic, e.g. a monoclonal antibody which distinguishes between the apo and holo forms of cytochrome *c* in solution combines equally well with both proteins on the solid phase. Covalent coupling to carboxy-derivatized plastic or capture of the antigen substrate by solid-phase antibody can sometimes lessen this effect.

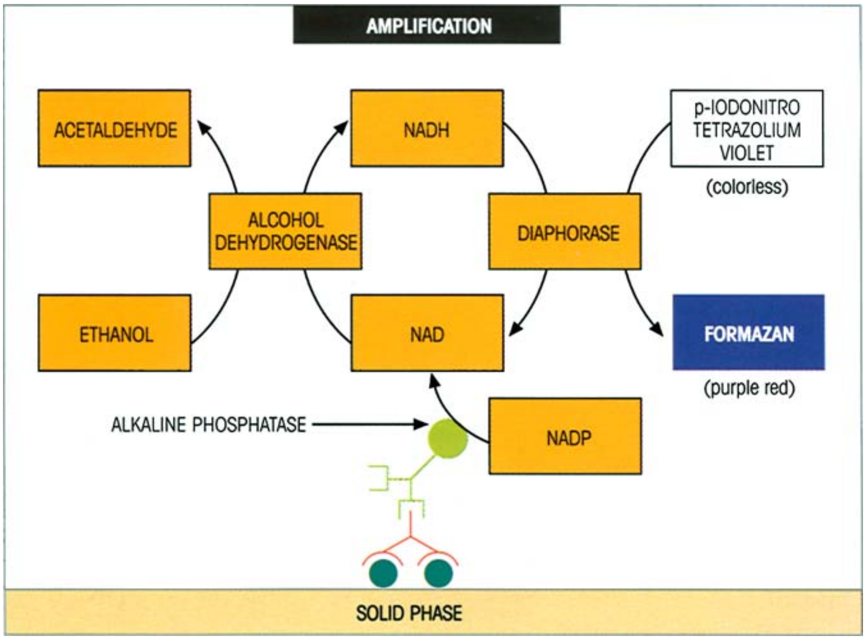


Figure 6.11. Coenzyme-geared amplification of the phosphatase reaction to reveal solid-phase anti-immunoglobulin label.

paper disk, which is then treated with patient's serum. The amount of specific IgE bound to the paper can now be estimated by the addition of labeled anti-IgE.

A wide variety of labels are available

Whilst providing extremely good sensitivity, radiolabels have a number of disadvantages, including loss of sensitivity during storage due to radioactive decay, the deterioration of the labeled reagent through radiation damage, and the precautions needed to minimize human exposure to radioactivity. Therefore, other types of label are often employed in immunoassays.

ELISA (enzyme-linked immunosorbent assay). Enzymes which give a colored soluble reaction product are currently the most commonly used labels, with horseradish peroxidase (HRP) and calf intestine alkaline phosphatase (AP) being by far the most popular. *Aspergillus niger* glucose oxidase, soy bean urease and *Escherichia coli* β-galactosidase provide further alternatives. One clever ploy for amplifying the phosphatase reaction is to use nicotinamide adenine dinucleotide phosphate (NADP) as a substrate to generate NAD which now acts as a coenzyme for a second enzyme system (figure 6.11).

Other labels. Enzyme-labeled streptococcal protein G or staphylococcal protein A will bind to IgG. Conjugation with the vitamin biotin is frequently used since this can readily be detected by its reaction with enzyme-linked avidin or streptavidin (the latter gives

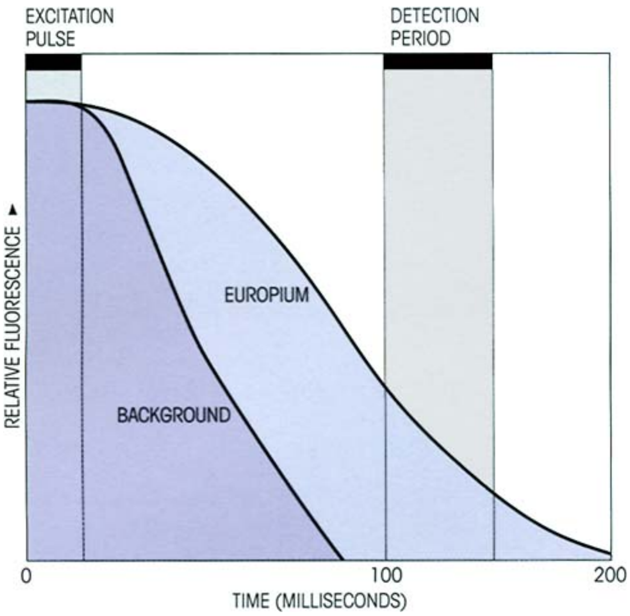


Figure 6.12. The principle of time-resolved fluorescence assay. The problem with conventional methods of detection of low fluorescent signals is interference from reflection of incident light and background instrument fluorescence. By using just a short excitation pulse and measuring the signal after background has fallen to zero but before the europium with its long fluorescence half-life has decayed completely, good discrimination between a weak signal and background becomes possible.

lower background binding), both of which bind with ferocious specificity and affinity ($K = 10^{15}\,\text{M}^{-1}$).

Chemiluminescent systems based on the HRP-catalysed enhanced luminol reaction, where light from the oxidized luminol substrate is intensified and the signal duration increased by the use of an enhancing reagent, provide increased sensitivity and dynamic range. Special mention should be made of time-resolved fluorescence assays based upon chelates of rare earths such as europium 3^+ (figure 6.12), although these have a more important role in antigen assays.

DETECTION OF IMMUNE COMPLEX FORMATION

Many techniques for detecting circulating complexes have been described and because of variations in the size, complement-fixing ability and Ig class of different complexes, it is useful to apply more than one method. Two fairly robust methods for general use are:

1 precipitation of complexed IgG from serum at concentrations of polyethylene glycol which do not bring down significant amounts of IgG monomer, followed by estimation of IgG in the precipitate by single radial immunodiffusion (SRID) or laser nephelometry, and

2 binding of C3b-containing complexes to beads coated with bovine conglutinin (cf. p. 17) and estimation of the bound Ig with enzyme-labeled anti-Ig.

Other techniques include: (i) estimation of the binding of ^{125}I-C1q to complexes by coprecipitation with polyethylene glycol, (ii) inhibition by complexes of rheumatoid factor-induced aggregation of IgG-coated particles, and (iii) detection with radiolabeled anti-Ig of serum complexes capable of binding to the C3b (and to a lesser extent the Fc) receptors on the Raji cell line. Sera from patients with immune complex disease often form a cryoprecipitate when allowed to stand at 4°C. Measurement of serum C3 and its conversion product C3c is sometimes useful.

Tissue-bound complexes are usually visualized by the immunofluorescent staining of biopsies with conjugated anti-immunoglobulins and anti-C3 (cf. figure 16.17).

IDENTIFICATION AND MEASUREMENT OF ANTIGEN

Precipitation reaction can be carried out in gels

Characterization of antigens by electrophoresis and immunofixation

This technique is most often applied to the detection of an abnormal protein in serum or urine, usually a monoclonal paraprotein secreted by a B-cell tumor. The paraprotein localizes as a dense compact 'M' band of defined electrophoretic mobility and its antigenic identity is then revealed by immunofixation with specific precipitating antiserums applied in paper strips overlying parallel lanes in the electrophoresis gel (figure 6.13).

Quantification by single radial immunodiffusion (SRID)

When antigen diffuses from a well into agar containing suitably diluted antiserum, initially it is present in a relatively high concentration and forms soluble complexes; as the antigen diffuses further the concentration continuously falls until the point is reached at which the reactants are nearer optimal proportions and a ring of precipitate is formed. The higher the concentration of antigen, the greater the diameter of this ring (figure 6.14). By incorporating, say, three standards of known antigen concentration in the plate, a calibration curve can be obtained and used to determine the amount of antigen in the unknown samples tested (figure 6.15). The method was used routinely in

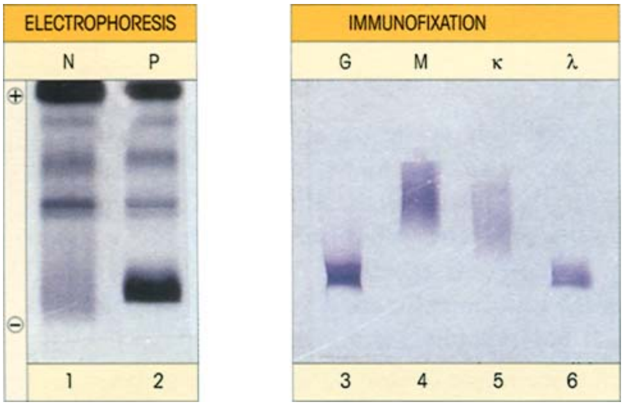


Figure 6.13. Electrophoresis and immunofixation of a paraprotein in serum. The sample is separated into its component bands by electrophoresis in agarose gel and these are visualized by direct staining after drying down the gel. The test sample is also run on a parallel gel which is then overlaid with strips of paper soaked in a specific antiserum. The antibodies diffuse into the gel and 'fix' the antigen by precipitation; after washing to remove the nonprecipitated soluble proteins, the gel is dried and stained to reveal the location of the paraprotein–antibody complex. N, normal serum; P, patient's serum showing compact paraprotein band; G, M, κ and λ represent immunofixations with antiserum specific for each immunoglobulin chain. In the example chosen, the paraprotein is an IgGλ. (Material kindly supplied by Mr T. Heys.)

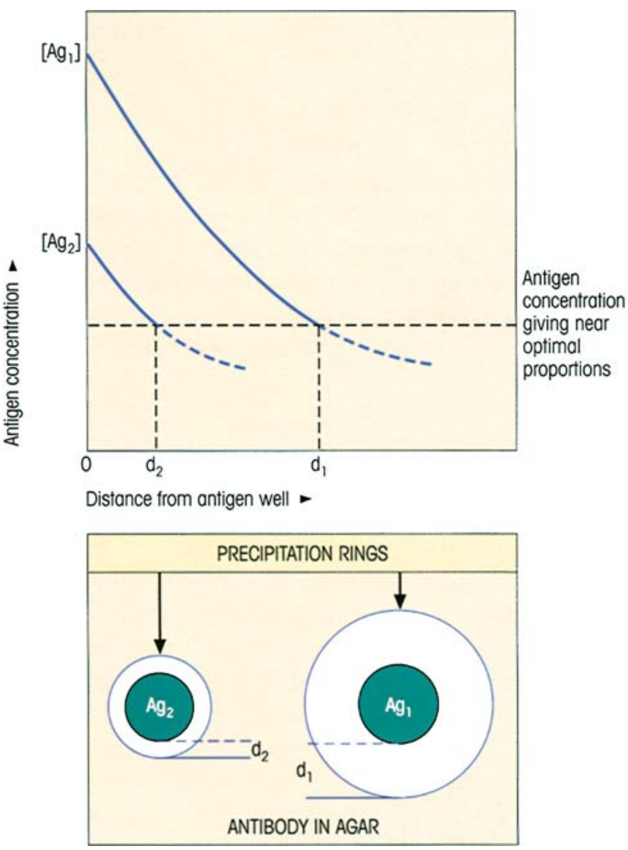


Figure 6.14. Single radial immunodiffusion: relation of antigen concentration to size of precipitation ring formed. Antigen at the higher concentration [Ag_1] diffuses further from the well before it falls to the level giving precipitation with antibody near optimal proportions.

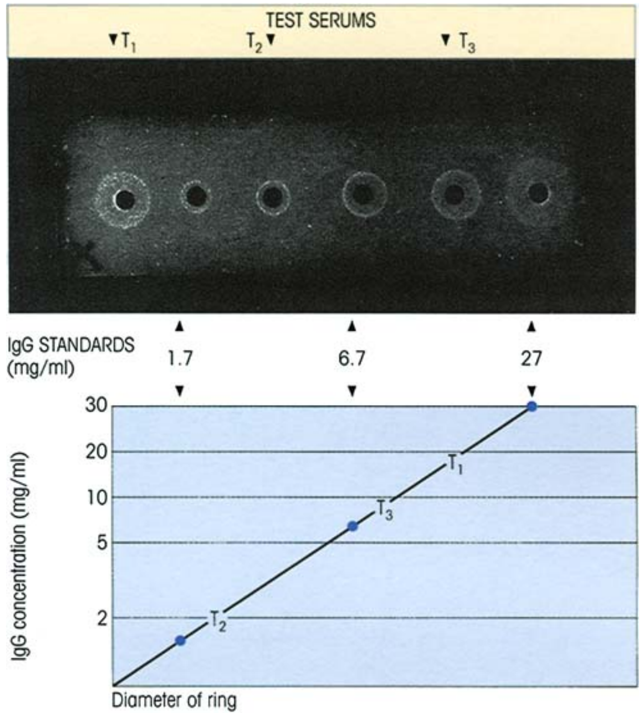


Figure 6.15. Measurement of IgG concentration in serum by single radial immunodiffusion. The diameter of the standards (●) enables a calibration curve to be drawn and the concentration of IgG in the serum under test can be read off:
T_1—serum from patient with IgG myeloma; 15 mg/ml;
T_2—serum from patient with hypogammaglobulinemia; 2.6 mg/ml;
T_3—normal serum; 9.6 mg/ml.
(Courtesy of Professor F.C. Hay.)

clinical immunology, particularly for immunoglobulin determinations, and also for substances such as the third component of complement, transferrin, C-reactive protein (CRP) and the embryonic protein, α-fetoprotein, which is associated with certain liver tumors. More affluent laboratories now tend to use nephelometry (see below).

The nephelometric assay for antigen

If antigen is added to a solution of excess antibody, the amount of complex which can be assessed by forward light scatter in a nephelometer (cf. p. 110) is linearly related to the concentration of antigen. With the ready availability of a wide range of monoclonal antibodies which facilitate the standardization of the method, nephelometry is replacing SRID for the estimation of immunoglobulins, C3, C4, haptoglobin, ceruloplasmin and CRP in those favored laboratories which can sport the appropriate equipment. Very small samples down in the range 1–10 μl can be analysed. Turbidity of the sample can be a problem; blanks lacking antibody can be deducted but a more satisfactory solution is to

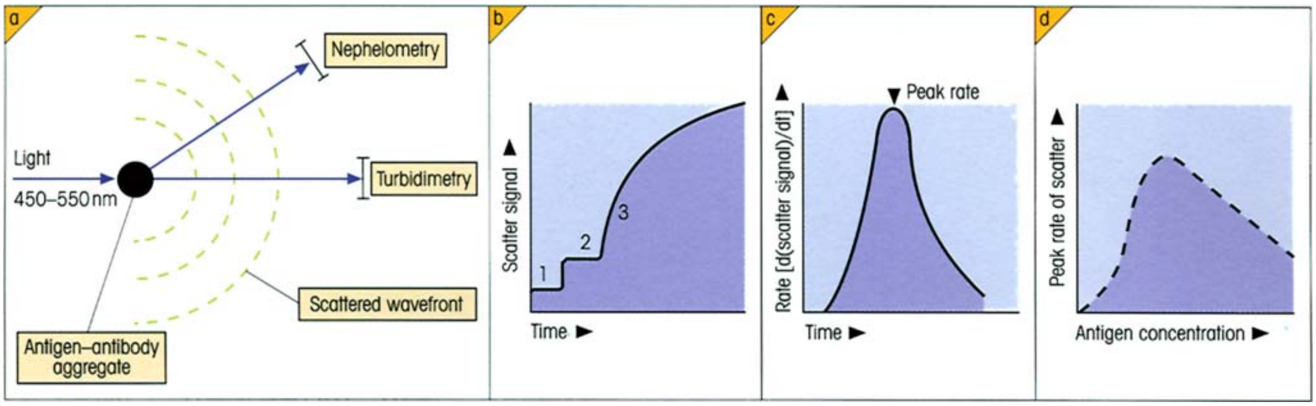


Figure 6.16. Rate nephelometry. (a) On addition of antiserum, small antigen–antibody aggregates form (cf. figure 6.2) which scatter incident light filtered to give a wavelength band of 450–550 nm. For nephelometry, the light scattered at a forward angle of 70° or so is measured. (b) After addition of the sample (1) and then the antibody (2), the rate at which the aggregates form (3) is determined from the scatter signal. (c) The software in the instrument then computes the maximum rate of light scatter which is related to the antigen concentration as shown in (d). (Copied from the operating manual for the 'Array' rate reaction automated immunonephelometer with permission from Beckman Coulter Ltd.)

follow the **rate of formation** of complexes which is proportional to antigen concentration since this obviates the need for a separate blank (figure 6.16). Because soluble complexes begin to be formed in antigen excess, it is important to ensure that the value for antigen was obtained in antibody excess by running a further control in which additional antigen is included.

Sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS–PAGE) for analysis of immunoprecipitates and immunoblotting

When proteins are denatured by SDS, the larger they are the more SDS they bind and the more negatively charged they become. Thus proteins of different size can be separated by electrophoresis in a gel such as polyacrylamide on the basis of their overall charge.

If one or more antigens are radiolabeled and the complex with added antibody is precipitated with an anti-Ig reagent such as staphylococci bearing protein A, SDS–PAGE of the complex followed by autoradiography should define the **number** and **molecular weights of the antigens** concerned (figure 6.17).

A **Western blot (immunoblot)** of antigens separated from a complex mixture by SDS–PAGE (or by isoelectric focusing) can be carried out. This employs transverse electrophoresis on to polyvinylidene difluoride (PVDF) or nitrocellulose membranes, where the proteins bind nonspecifically and are subsequently identified by staining with appropriately labeled antibodies. Obviously, such a procedure will not work with antigens which are irreversibly denatured by this detergent, and it is best to use polyclonal antisera for blotting to increase the chance of including antibodies to whichever epitopes do survive the denaturation procedure; a surprising number do (figure 6.18). Conversely, the spectrotype of an antiserum can be revealed by isoelectric focusing, blotting and then staining with labeled antigen.

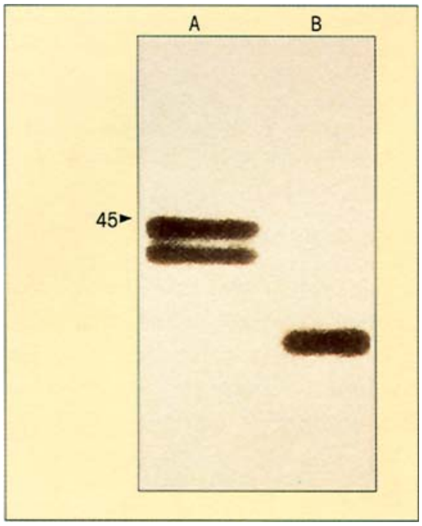


Figure 6.17. Immunoprecipitation of membrane antigen. Analysis of membrane-bound class I MHC antigens (cf. p. 70). The membranes from human cells pulsed with ^{35}S-methionine were solubilized in a detergent, mixed with a monoclonal antibody to HLA-A and B molecules and immunoprecipitated with staphylococci. An autoradiograph (A) of the precipitate run in SDS–PAGE shows the HLA-A and B chains as a 43 000 molecular weight doublet (the position of a 45 000 marker is arrowed). If membrane vesicles are first digested with proteinase K before solubilization, a labeled band of molecular weight 39 000 can be detected (B) consistent with a transmembrane orientation of the HLA chain: the 4000 Da hydrophilic C-terminal fragment extends into the cytoplasm and the major portion, recognized by the monoclonal antibody and by tissue typing reagents, is present on the cell surface (cf. figure 4.11). (From data and autoradiographs kindly supplied by Dr M.J. Owen.)

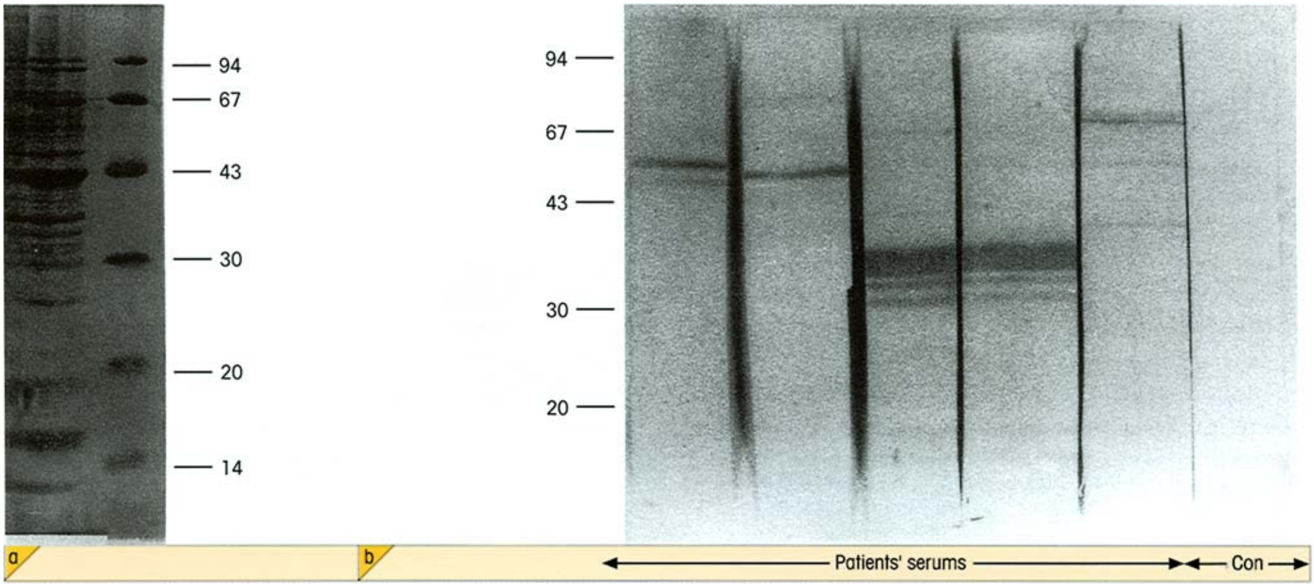


Figure 6.18. Western blot analysis of human polymorph primary granules with sera from patients with systemic vasculitis. Human polymorph postnuclear supernatant was run on SDS–PAGE. (a) Gel stained for protein with Coomassie Blue. Numbers refer to molecular weight markers (kDa). (b) Blots from gel stained with serums from five patients and one control and visualized with alkaline phosphatase-conjugated goat anti-human IgG. Three different patterns of autoantibody reaction are evident. (Kindly supplied by Drs J. Cambridge & B. Leaker.) The antibodies used for immunoblotting are usually labeled with enzymes or biotin (followed by enzyme-labeled avidin).

The immunoassay of antigens

The ability to establish the concentration of an analyte (i.e. a substance to be measured) through fractional occupancy of its specific binding reagent is a feature of any ligand-binding system (Milestone 6.1), but because antibodies can be raised to virtually any structure, its application is most versatile in immunoassay.

Large analytes, such as protein hormones, are usually estimated by a noncompetitive two-site assay in which the original ligand binder and the labeled detection reagent are both antibodies (figure M6.1.1). By using monoclonal antibodies directed to two different epitopes on the same analyte, the system has greater power to discriminate between two related analytes; if the fractional cross-reactivity of the first antibody for a related analyte is 0.1 and of the second also 0.1, the final readout for cross-reactivity will be as low as 0.1×0.1, i.e. 1%. Using chemiluminescent and time-resolved fluorescent probes, highly sensitive assays are available for an astonishing range of analytes.

For small molecules like drugs or steroid hormones, where two-site binding is impractical, competitive assays (figure M6.1.1) are appropriate.

Immunoassay on multiple microspots

Paradoxically, minute spots of solid-phase antibody are not completely saturated by analyte in the range of concentrations normally worked with in most immunoassays. Rather, analysis has shown that the fractional occupancies expected permit immunoassay technology, or any other ligand-specific binding system, to be practical options (Ekins), particularly with the advent of modern highly sensitive probes. Furthermore, when the amount of antibody on the microspot is very small, the fractional occupancy is independent of antibody level and also of analyte volume (**the ambient analyte principle**). Sensitivities compare very favorably with the best immunoassays and, with such miniaturization, arrays of microspots which capture antibodies of different specificities can be placed on a single chip, opening the door to multiple analyte screening in a single test, with each analyte being identified by its grid coordinates in the array.

Epitope mapping

T-cell epitopes

Where the primary sequence of the whole protein is known, the identification of T-cell epitopes is comparatively straightforward. Since these epitopes are linear in nature, multipin solid-phase synthesis can be employed to generate a series of overlapping peptides, 8–9-mers for cytotoxic T-cells and usually 10–14-mers for T-helpers (figure 6.19), and their ability to react

Milestone 6.1 — Ligand-binding Assays

The appreciation that a ligand could be measured by the fractional occupancy (F) of its specific binding agent heralded a new order of sensitive wide-ranging assays. Ligand-binding assays were first introduced for the measurement of thyroid hormone by thyroxine-binding protein (Ekins) and for the estimation of hormones by antibody (Berson & Yalow). These findings spawned the technology of radioimmunoassay, so called because the antigen had to be trace-labeled in some way and the most convenient candidates for this were radioisotopes.

The relationship between fractional occupancy and analyte concentration [An] is given by the equation:

$$F = 1 - (1/1 + K[\text{An}])$$

where K is the association constant of the ligand-binding reaction. F can be measured by noncompetitive or competitive assays (figure M6.1.1) and related to a calibration curve constructed with standard amounts of analyte.

For competitive assays, the maximum theoretical sensitivity is given by the term ε/K where ε is the experimental error (coefficient of variation). Suppose the error is 1% and K is $10^{11}\,\text{M}^{-1}$, the maximum sensitivity will be $0.01 \times 10^{-11}\,\text{M} = 10^{-13}\,\text{M}$ or 6×10^{7} molecules/ml. For noncompetitive assays, labels of very high specific activity could give sensitivities down to 10^{2}–10^{3} molecules/ml under ideal conditions. In practice, however, since the sensitivity represents the lowest analyte concentration which can be measured against a background containing zero analyte, the error of the measurement of background poses an ultimate constraint on sensitivity.

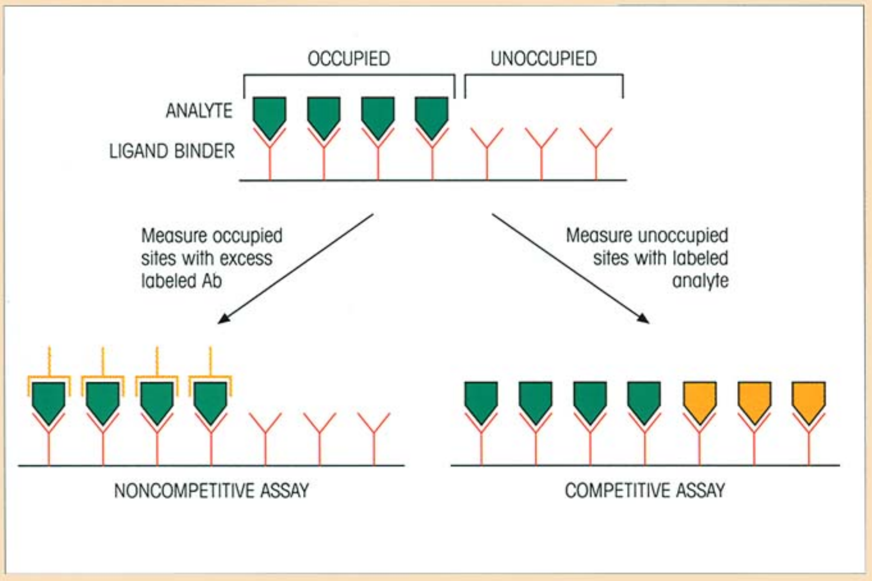


Figure M6.1.1. The principle of ligand-binding assays. The ligand-binding agent may be in the soluble phase or bound to a solid support as shown here, the advantage of the latter being the ease of separation of bound from free analyte. After exposure to analyte, the fractional occupancy of the ligand-binding sites can be determined by competitive or noncompetitive assays using labeled reagents (in orange) as shown.

with antigen-specific T-cell lines or clones can be deciphered to characterize the active epitopes.

Dissecting out T-cell epitopes where the antigen has not been characterized is a more daunting task. Randomized peptide libraries can be produced but strategies need to be devised in order to keep these within manageable numbers. Information from the accumulated data deposited in various databanks can be used to identify key anchor residues and libraries constructed that maintain the relevant amino acids at these positions. Thus, a positional scanning approach employs a peptide library in which one amino acid at a particular position is kept constant and all the different amino acids are used at the other positions.

B-cell epitopes

If they are linear protein epitopes formed directly from the primary amino acid sequence, then binding of antibody to individual overlapping peptides synthesized

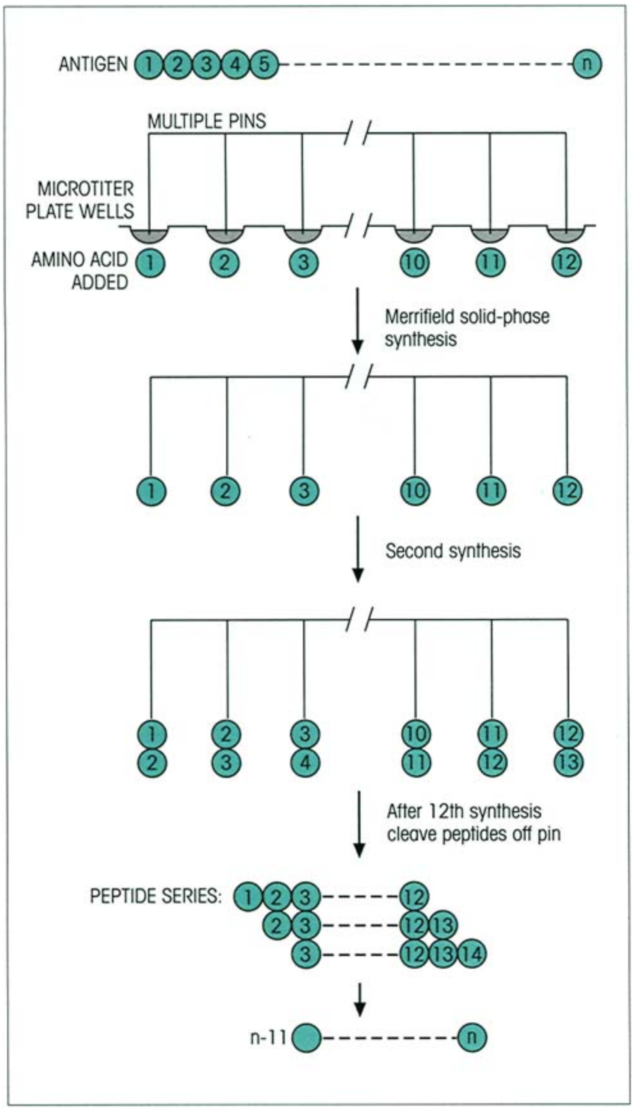


Figure 6.19. Synthesis of overlapping peptide sequences for (PEPSCAN) epitope analysis. A series of pins which sit individually in the wells of a 96-well microtiter plate each provide a site for solid-phase synthesis of peptide. A sequence of such syntheses as shown in the figure provides the required nests of peptides. Incorporation of a readily cleavable linkage allows the soluble peptide to be released as the synthesis is terminated.

as described above will identify them. Unfortunately, most epitopes on globular proteins recognized by antibody are discontinuous and this makes the job rather demanding, since one cannot predict which residues are likely to be brought together in space to form the epitope. To the extent that small linear sequences may contribute to a discontinuous epitope, the overlapping peptide strategy may provide some clues.

A potentially promising approach to this problem of mimicking the residues which constitute such epitopes (termed **mimotopes** by Geysen) is through the production of libraries of bacteriophages bearing all possible random hexapeptides. These are produced by ligating degenerate oligonucleotide inserts (coding for hexapeptides) to a bacteriophage coat protein in a suitable vector; appropriate expression in *E. coli* can provide up to 10^9 different clones. The beauty of the system is that a bacteriophage expressing a given hexapeptide on its external coat protein also bears the sequence encoding the hexapeptide in its genome (cf. p. 124). Accordingly, sequential rounds of selection, in which the phages react with a biotinylated monoclonal antibody and are then panned on a streptavidin plate, should isolate those bearing the peptides which mimic the epitope recognized by the monoclonal; nucleotide sequencing will then give the peptide structure.

Even nonproteinaceous antigens can occasionally be mimicked using peptide libraries, one example being the use of a D-amino acid hexapeptide library to identify a mimotope for *N*-acetylglucosamine. Others have used a single-chain Fv (scFv) library to isolate an idiotypic mimic of a meningococcal carbohydrate.

MAKING ANTIBODIES TO ORDER

The monoclonal antibody revolution

First in rodents

A fantastic technological breakthrough was achieved by Köhler and Milstein who devised a technique for the production of 'immortal' clones of cells making single antibody specificities by fusing normal antibody-forming cells with an appropriate B-cell tumor line. These so-called 'hybridomas' are selected out in a tissue culture medium which fails to support growth of the parental cell types, and by successive dilutions or by plating out, single clones can be established (figure 6.20). These clones can be grown up in the ascitic form in mice when quite prodigious titers of monoclonal antibody can be attained, but bearing in mind the imperative to avoid using animals wherever feasible, propagation in large-scale culture is to be preferred. Remember that, even in a good antiserum, over 90% of the Ig molecules have little or no avidity for the antigen, and the 'specific antibodies' themselves represent a whole spectrum of molecules with different avidities directed against different determinants on the antigen. What a contrast is provided by the monoclonal antibodies, where all the molecules produced by a given hybridoma are identical: they have the same Ig class and allotype, the same variable region, structure, idiotype, affinity and specificity for a given epitope.

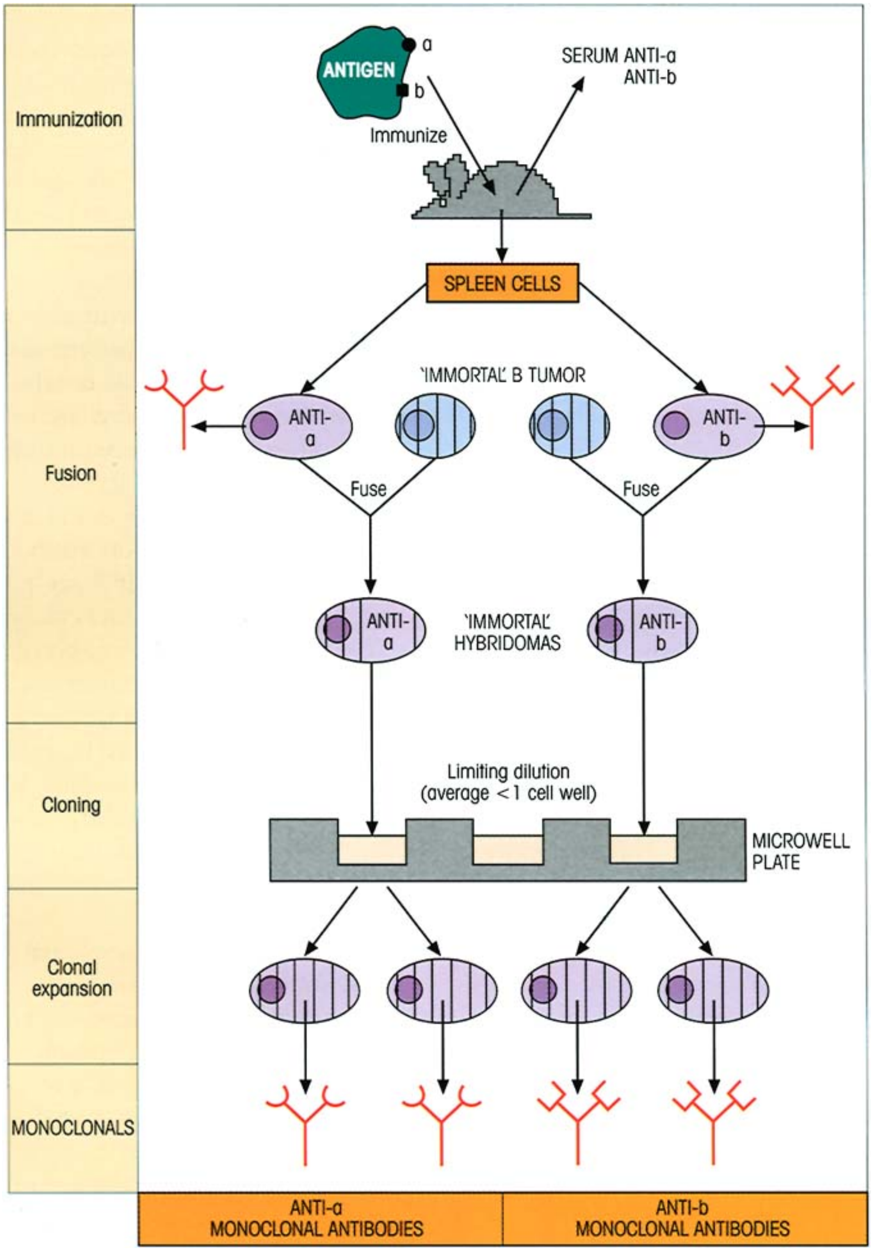


Figure 6.20. Production of monoclonal antibodies. Mice immunized with an antigen bearing (shall we say) two epitopes, a and b, develop spleen cells making anti-a and anti-b which appear as antibodies in the serum. The spleen is removed and the individual cells are fused in polyethylene glycol with constantly dividing (i.e. 'immortal') B-tumor cells selected for a purine enzyme deficiency and usually for their inability to secrete Ig. The resulting cells are distributed into microwell plates in HAT (hypoxanthine, aminopterin, thymidine) medium which kills off the fusion partners. They are seeded at such a dilution that on average each well will contain less than one hybridoma cell. Each hybridoma—the fusion product of a single antibody-forming cell and a tumor cell—will have the ability of the former to secrete a single species of antibody and the immortality of the latter enabling it to proliferate continuously. Thus, clonal progeny can provide an unending supply of antibody with a single specificity—the monoclonal antibody. In this example, we considered the production of hybridomas with specificity for just two epitopes, but the same technique enables monoclonal antibodies to be raised against complex mixtures of multiepitopic antigens. Fusions using rat cells instead of mouse may have certain advantages in giving a higher proportion of stable hybridomas, and monoclonals which are better at fixing human complement, a useful attribute in the context of therapeutic applications to humans involving cell depletion.

Naturally, for use in the human, the ideal solution is the production of purely human monoclonals. Human myeloma fusion partners have not found wide acceptance since they tend to have low fusion efficiencies, poor growth and secretion of the myeloma Ig which dilutes the desired monoclonal. A nonsecreting heterohybridoma obtained by fusing a mouse myeloma with human B-cells can be used as a productive fusion partner for antibody-producing human B-cells. Other groups have turned to the well-characterized murine fusion partners, and the heterohybridomas so formed grow well, clone easily and are productive. There is some instability from chromosome loss and it appears that antibody production is maintained by translocation of human *Ig* genes to mouse chromosomes. Fusion frequency is even better if Epstein–Barr virus (EBV)-transformed lines are used instead of B-cells.

Whereas the large amount of nonspecific, relative to antigen-specific, Ig in an antiserum means that background binding to antigen in any given immunological test may be uncomfortably high, the problem is greatly reduced with a monoclonal antibody preparation, since all the Ig is antibody, thus giving a much superior 'signal: noise' ratio. By being directed towards single epitopes on the antigen, monoclonal antibodies frequently show high specificity in terms of their low cross-reactivity with other antigens. Occasionally, however, one sees quite unexpected binding to molecules which react poorly, if at all, with a specific antiserum directed to the original antigen. The reason for this has already been discussed (see p. 89). Suffice it to say here that the problem can be circumvented by using a group of overlapping monoclonals reacting with the same determinant or a combination of monoclonals to more than one determinant on the same antigen.

An outstanding advantage of the monoclonal antibody as a reagent is that it provides a single standard material for all laboratories throughout the world to use in an unending supply if the immortality and purity of the cell line are nurtured; antisera raised in different animals, on the other hand, may be as different from each other as chalk and cheese. The monoclonal approach again shows a clean pair of heels relative to conventional strategies in the production of antibodies specific for individual components in a complex mixture of antigens. The uses of monoclonal antibodies are truly legion and include: immunoassay, diagnosis of malignancies, tissue typing, serotyping of microorganisms, the separation of individual cell types with specific surface markers (e.g. lymphocyte subpopulations), therapeutic neutralization of inflammatory cytokines and 'magic bullet' therapy with cytotoxic agents coupled to antitumor-specific antibody—these and many other areas have been transformed by hybridoma technology.

Catalytic antibodies

An especially interesting development with tremendous potential is the recognition that a monoclonal antibody to a stable analog of the transition state of a given reaction can act as an enzyme ('abzyme') in catalysing that reaction. The possibility of generating enzymes to order promises a very attractive future, and some exceedingly adroit chemical maneuvers have already extended the range of reactions which can be catalysed in this way. A recent demonstration of sequence-specific peptide cleavage with an antibody which incorporates a metal complex cofactor has raised the pulse rate of the *cognoscenti*, since this is an energetically difficult reaction which has an enormous range of applications. Another innovative approach is to immunize with an antigen which is so highly reactive that a chemical reaction occurs in the antibody combining site. This recruits antibodies which are not only complementary to the active chemical, but are also likely to have some enzymic power over the immunogen–substrate complex. Thus, using this strategy, an antibody with exceptionally broad substrate specificity for efficient catalysis of aldol and retro-aldol reactions was obtained. A key feature of this antibody is a reactive lysine buried within a hydrophobic pocket in the binding site. The antibody remains catalytically active for several weeks following i.v. injection into mice and has therapeutic potential for a version of antibody-directed enzyme prodrug therapy (ADEPT, see p. 392), here with the enzyme component being a catalytic antibody.

Large combinatorial antibody libraries created by random association between pools of heavy and light chains and expressed on bacteriophages (see below) can be screened for catalytic antibodies by using the substrate in a solid-phase state. Cleavage by the catalytic antibody leaves a solid-phase product which can now be identified by a double antibody system using antibodies specific for the product as distinct from the substrate.

An area of great interest is the presence of catalytic autoantibodies in certain groups of patients, with hydrolytic antibodies against vasoactive intestinal peptide, DNA and thyroglobulin having been described. Catalytic antibodies capable of factor VIII hydrolysis have also recently been discovered in hemophiliacs given this clotting factor, the antibodies preventing the coagulation function of the factor VIII.

Human monoclonals can be made

Mouse monoclonals injected into human subjects for therapeutic purposes are frightfully immunogenic and the human anti-mouse antibodies (HAMA in the trade) so formed are a wretched nuisance, accelerating clearance of the monoclonal from the blood and possibly causing hypersensitivity reactions; they also prevent the mouse antibody from reaching its target and, in some cases, block its binding to antigen. In some circumstances it is conceivable that a mouse monoclonal taken up by a tumor cell could be processed and become the MHC-linked target of cytotoxic T-cells or help to boost the response to a weakly immunogenic antigen on the tumor cell surface. In general, however, logic points to removal of the xenogeneic (foreign) portions of the monoclonal antibody and their replacement by human Ig structures using recombinant DNA technology. Chimeric constructs, in which the V_H and V_L mouse domains are spliced onto human C_H and C_L genes (figure 6.21a), are far less immunogenic in humans.

A more refined approach is to graft the six complementarity determining regions (CDRs) of a high affinity rodent monoclonal onto a completely human Ig framework without loss of specific reactivity (figure 6.21b). This is not a trivial exercise, however, and the objective of fusing human B-cells to make hybridomas is still appealing, taking into account not only the gross reduction in immunogenicity, but also the fact that, within a species, antibodies can be made to subtle differences such as major histocompatibility complex (MHC) polymorphic molecules and tumor-associated antigens on other individuals. In contrast, xenogeneic responses are more directed to immunodominant structures common to most subjects, making the production of variant-specific antibodies more difficult. Notwithstanding the difficulties in finding good fusion partners, large numbers of human monoclonals have been established. A further restriction arises because the peripheral blood B-cells, which are the only B-cells readily available in the human, are not normally regarded as a good source of antibody-forming cells.

Immortalized Epstein–Barr virus-transformed B-cell lines have also been used as a source of human monoclonal antibodies. Although these often produce relatively low affinity IgM antibodies, some useful higher affinity IgG antibodies can occasionally be obtained. The cell lines frequently lose their ability to secrete antibody if cultured for long periods of time, although they can sometimes be rescued by fusion with a myeloma cell line to produce hybridomas, or the genes can be isolated and used to produce a recombinant antibody.

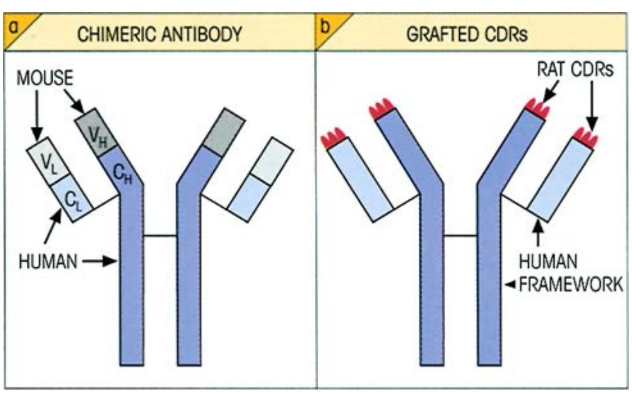


Figure 6.21. Genetically engineering rodent antibody specificities into the human. (a) Chimeric antibody with mouse variable regions fused to human Ig constant regions. (b) 'Humanized' rat monoclonal in which gene segments coding for all six CDRs are grafted onto a human Ig framework.

A radically different approach involves the production of transgenic xenomouse strains in which megabase-sized unrearranged human Ig *H* and *κ* light chain loci have been introduced into mice whose endogenous murine *Ig* genes have been inactivated. Immunization of these mice yields high affinity (10^{-10}–10^{-11} M) human antibodies which can then be isolated using hybridoma or recombinant approaches. Potent anti-inflammatory (anti-IL-8) and anti-tumor (anti-epidermal growth factor receptor) therapeutic agents have already been obtained using such mice.

There is still a snag in that even human antibodies can provoke anti-idiotype responses; these may have to be circumvented by using engineered antibodies bearing different idiotypes for subsequent injections. Even more desirable would be if the prospective recipients could be first made tolerant to the idiotype, perhaps by coadministering the therapeutic antibody together with a nondepleting anti-CD4.

Many human monoclonals are awaiting the go-ahead for clinical use; one can cite IgG anti-RhD for the prevention of rhesus disease of the newborn (see p. 334), and highly potent monoclonals for protection against varicella zoster, cytomegalovirus, group B streptococci and lipopolysaccharide endotoxins of Gram-negative bacteria.

Engineering antibodies

There are other ways around the problems associated with the production of human monoclonals which exploit the wiles of modern molecular biology. Reference has already been made to the 'humanizing' of rodent antibodies (figure 6.21), but an important new strategy based upon bacteriophage expression and **selection** has achieved a prominent position. In essence, mRNA from primed human B-cells is converted to cDNA and the antibody genes, or fragments therefrom, expanded by the polymerase chain reaction (PCR). Single constructs are then made in which the light and heavy chain genes are allowed to combine randomly in tandem with the gene encoding bacteriophage coat protein III (pIII) (figure 6.22). This **combinatorial library** containing most random pairings of heavy and light chain genes encodes a huge repertoire of antibodies (or their fragments) expressed as fusion proteins with pIII on the bacteriophage surface. The extremely high number of phages produced by *E. coli* infection can now be panned on solid-phase antigen to select those bearing the highest affinity antibodies attached to their

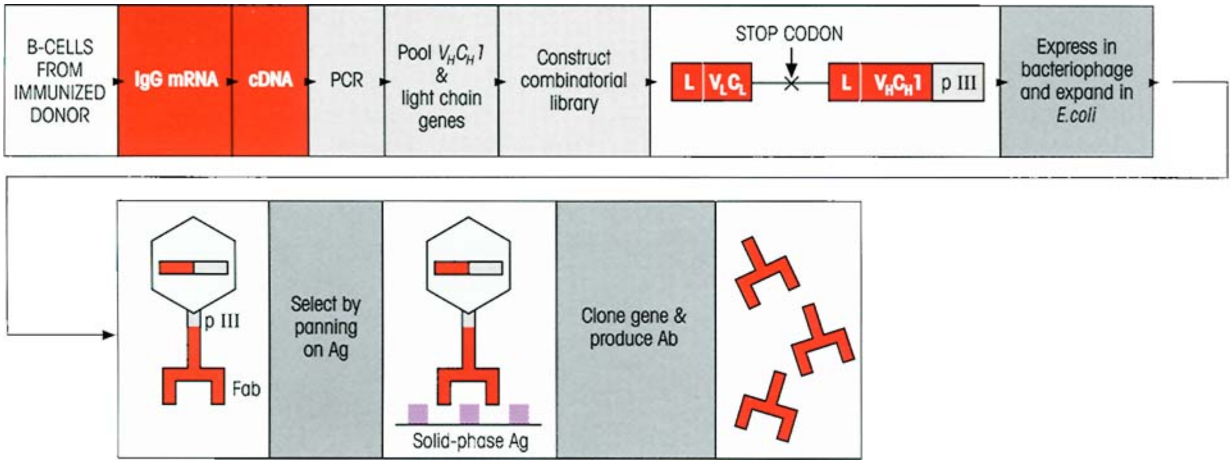


Figure 6.22. Selection of antibody genes from a combinatorial library. B-cells from an immunized donor (in one important experiment, human memory peripheral blood cells were boosted with tetanus toxoid antigen after transfer to SCID mice; Duchosal M.A. *et al.* (1992) *Nature* **355**, 258) are used for the extraction of IgG mRNA and the light chain (V_LC_L) and V_HC_H1 genes (encoding Fab) randomly combined in constructs fused to the bacteriophage pIII coat protein gene as shown. These were incorporated into phagemids such as pHEN1 and expanded in *E. coli*. After infection with helper phage, the recombinant phages bearing the highest affinity were selected by rounds of panning on solid-phase antigen so that the genes encoding the Fabs could be cloned. L = bacterial leader sequence.

surface (figure 6.22). Because the genes which encode these highest affinity antibodies are already present within the selected phage, they can readily be cloned and the antibody expressed in bulk. It should be recognized that this **selection** procedure has an enormous advantage over techniques which employ **screening** because the number of phages which can be examined is several logs higher.

Combinatorial libraries have also been established using mRNA from **unimmunized** human donors. V_H, V_κ and V_λ genes are expanded by PCR and randomly recombined to form single-chain Fv (scFv) constructs (figure 6.23a) fused to phage pIII. Soluble fragments binding to a variety of antigens have been obtained. Of special interest are those which are autoantibodies to molecules with therapeutic potential such as CD4 and tumor necrosis factor-α (TNFα); lymphocytes expressing such autoantibodies could not be obtained by normal immunization since they would probably be tolerized, but the random recombination of V_H and V_L can produce entirely new specificities under conditions *in vitro* where tolerance mechanisms do not operate.

Although a 'test-tube' operation, this approach to the generation of specific antibodies does resemble the affinity maturation of the immune response *in vivo* (see p. 195) in the sense that antigen is the determining factor in selecting out the highest affinity responders.

In order to increase the affinities of antibodies produced by these techniques, antigen can be used to select higher affinity mutants produced by random mutagenesis or even more effectively by site-directed replacements at mutational hotspots (figure 6.23b), again mimicking the natural immune response which involves random mutation and antigen selection (see p. 192). Affinity has also been improved by gene 'shuffling' in which a V_H gene encoding a reasonable affinity antibody is randomly combined with a pool of V_L genes and subjected to antigen selection. The process can be further extended by mixing the V_L from this combination with a pool of V_H genes. It has also proved possible to shuffle individual CDRs between variable regions of moderate affinity antibodies obtained by panning on antigen, thereby creating antibodies of high affinity from relatively small libraries.

Other novel antibodies have been created. In one construct, two scFv fragments associate to form an antibody with two different specificities (figure 6.23c). Another consists of a single heavy chain variable region domain (DAB) whose affinity can be surprisingly high—of the order of 20 nM. If it were possible to overcome the 'stickiness' of these miniantibodies, their small size could be exploited for tissue penetration. The design of potential 'magic bullets' for immunotherapy can be based on fusion of a toxin (e.g. ricin) to an antibody Fab (figure 6.23d).

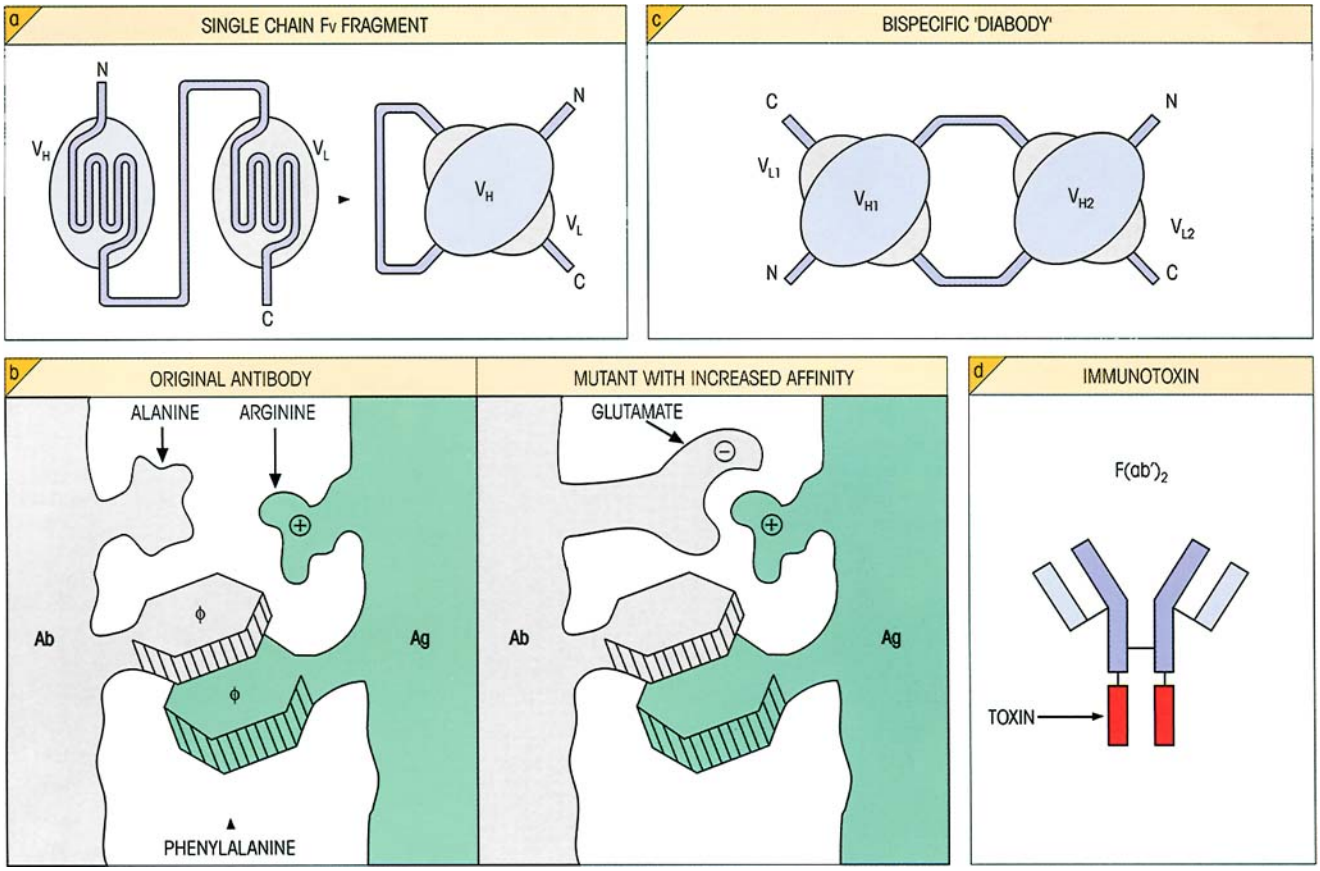


Figure 6.23. Other engineered antibodies. (a) A single gene encoding V_H and V_L joined by a sequence of suitable length gives rise to a single-chain Fv (scFv) antigen-binding fragment. (b) By site-specific mutagenesis of residues in or adjacent to the complementarity determining region (CDR), it is possible to increase the affinity of the antibody. (c) Two scFv constructs expressed simultaneously will associate to form a 'diabody' with two specificities. These bispecific antibodies have a number of uses. Note that such a bispecific antibody directed to two different epitopes on the same antigen will have a much higher affinity due to the 'bonus effect' of cooperation between the two binding sites (cf. p. 90). (d) Potential 'magic bullets' can be constructed by fusing the gene for a toxin (e.g. ricin) to the Fab.

Fields of antibodies

Not only can the genes for a monoclonal antibody be expressed in bulk in the milk of lactating animals but plants can also be exploited for this purpose. So-called '**plantibodies**' have been expressed in bananas, potatoes and tobacco plants. One can imagine a high-tech farmer drawing the attention of a bemused visitor to one field growing anti-tetanus toxoid, another anti-meningococcal polysaccharide, and so on. Multifunctional plants might be quite profitable with, say, the root being harvested as a food crop and the leaves expressing some desirable gene product. At this rate there may not be much left for science fiction authors to write about!

Drugs can be based on the CDRs of minibodies

Millions of **minibodies** composed of a segment of the V_H region containing three β-strands and the H1 and H2 hypervariable loops were generated by randomization of the CDRs and expressed on the bacteriophage pIII coat protein. By panning the library on functionally important ligand-binding sites, such as hormone receptors, useful lead candidates for drug design programs can be identified and their affinity improved by loop optimization, loop shuffling and further selection.

PURIFICATION OF ANTIGENS AND ANTIBODIES BY AFFINITY CHROMATOGRAPHY

The principle is simple and *very* widely applied. Antigen or antibody is bound through its free amino groups to cyanogen bromide-activated Sepharose particles. Insolubilized antibody, for example, can be used to pull the corresponding antigen out of solution, in

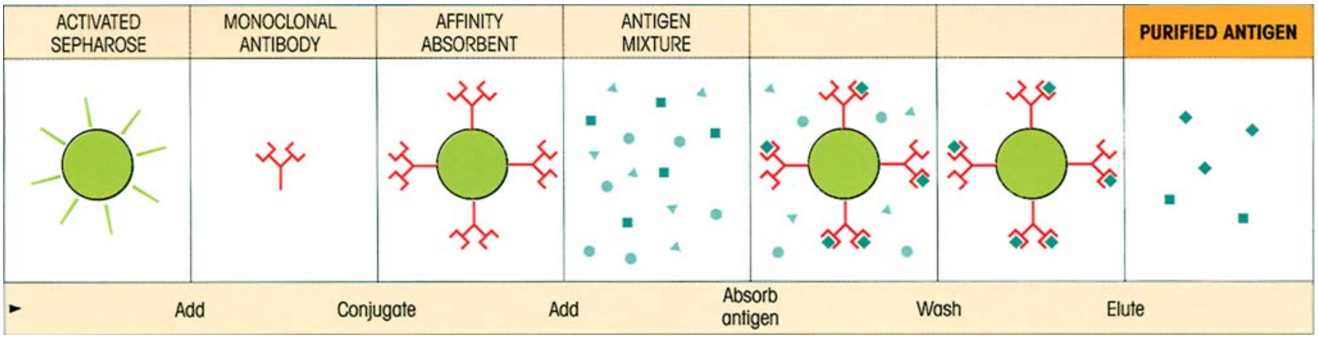


Figure 6.24. Affinity chromatography. A column is filled with Sepharose-linked antibody. The antigen mixture is poured down the column. Only the antigen binds and is released by a change in pH, for example. Conversely, an antigen-linked affinity column can be used to purify antibody.

which it is present as one component of a complex mixture, by absorption to its surface. The uninteresting garbage is washed away and the required ligand released from the affinity absorbent by disruption of the antigen–antibody bonds by changing the pH or adding chaotropic ions such as thiocyanate (figure 6.24). Likewise, an antigen immunosorbent can be used to absorb out an antibody from a mixture whence it can be purified by elution.

NEUTRALIZATION OF BIOLOGICAL ACTIVITY

To detect antibody

A number of biological reactions can be inhibited by addition of specific antibody. Thus the agglutination of red cells by interaction of influenza virus with receptors on the erythrocyte surface can be blocked by antiviral antibodies and this forms the basis for their serological detection. A test for antibodies to *Salmonella* H antigen present on the flagella depends upon their ability to inhibit the motility of the bacteria *in vitro*. Likewise, *Mycoplasma* antibodies can be demonstrated by their inhibitory effect on the metabolism of the organisms in culture.

Using antibody as an inhibitor

The successful treatment of cases of drug overdose with the Fab fragment of specific antibodies has been described and may become a practical proposition if a range of hybridomas can be assembled. Conjugates of cocaine with keyhole limpet hemocyanin can provoke neutralizing antibodies. Antibodies to hormones such as insulin and thyroid-stimulating hormone (TSH), or to cytokines, can be used to probe the specificity of biological reactions *in vitro*. For example, the specificity of the insulin-like activity of a serum sample on rat epididymal fat pad can be checked by the neutralizing effect of an antiserum. Such antibodies can be effective *in vivo*, and anti-TNF treatment of patients with rheumatoid arthritis has confirmed the role of this cytokine in the disease process. Likewise, as part of the worldwide effort to prevent disastrous overpopulation, attempts are in progress to immunize against chorionic gonadotropin using fragments of the β chain coupled to appropriate carriers, since this hormone is needed to sustain the implanted ovum.

In a totally different context, antibodies raised against myelin-associated neurite growth inhibitory proteins revealed their importance in preventing nerve repair, in that treatment with these antibodies permitted the regeneration of corticospinal axons after a spinal cord lesion had been induced in adult rats. This quite remarkable finding significantly advances our understanding of the processes involved in regeneration and gives ground for cautious optimism concerning the development of treatment for spinal cord damage, although for various reasons this may not ultimately be based on antibody therapy.

SUMMARY

Estimation of antibody

- The antibody content of a polyclonal antiserum is defined entirely in operational terms by the nature of the assay employed.
- Antibody in solution can be assayed by the formation of frank precipitates which can be enhanced by countercurrent electrophoresis in gels.
- Nonprecipitating antibodies can be measured by laser nephelometry or by salt or anti-Ig coprecipitation with radioactive antigen.
- Affinity is measured by a variety of methods including surface plasmon resonance which gives a measure of both the on- and off-rates.
- Antibodies can also be detected by macroscopic agglutination of antigen-coated particles, and by one of the most important methods, ELISA, a two-stage procedure in which antibody bound to solid-phase antigen is detected by an enzyme-linked anti-Ig.

Identification and measurement of antigen

- Antigens can be separated and their mobility in an electric field determined by electrophoresis and immunofixation.
- Antigens can be quantified by their reaction in gels with antibody using single radial immunodiffusion.
- Higher concentrations of antigens are frequently estimated by nephelometry.
- Exceedingly low concentrations of antigens can be measured by immunoassay techniques which depend upon the relationship between Ag concentration and fractional occupancy of the binding antibody. Occupied sites are measured with a high specific activity second antibody directed to a different epitope; alternatively, unoccupied sites can be estimated by labeled Ag. Multiple assays on arrays of antibody microspots are being developed.
- Overlapping nests of peptides derived from the linear sequence of a protein can map T-cell epitopes and the linear elements of B-cell epitopes. Bacteriophages encoding all possible hexapeptides on their surface have provided some limited success in identifying discontinuous B-cell determinants.

Detection of immune complexes

- Serum complexes can be determined by precipitation with polyethylene glycol, reaction with C1q or bovine conglutinin, changes in C3 and C3c and binding to rheumatoid factors.

Making antibodies to order

- Immortal hybridoma cell lines making monoclonal antibodies provide powerful immunological reagents and insights into the immune response. Applications include enumeration of lymphocyte subpopulations, cell depletion, immunoassay, cancer diagnosis and imaging, purification of antigen from complex mixtures, and recently the use of monoclonals as artificial enzymes (catalytic antibodies).
- Genetically engineered human antibody fragments can be derived by expanding the V_H and V_L genes from unimmunized, but preferably immunized, donors and expressing them as completely randomized combinatorial libraries on the surface of bacteriophage. Phages bearing the highest affinity antibodies are selected by panning on antigens and the antibody genes can then be cloned from the isolated viruses.
- Single-chain Fv (scFv) fragments encoded by linked V_H and V_L genes and even single heavy chain domains can be created.
- The HAMA response against mouse monoclonal antibodies can be reduced by producing chimeric antibodies with mouse variable regions and human constant regions or, better still, using humanized antibodies in which all the mouse sequences except for the CDRs are replaced by human sequences.
- Transgenic mice bearing human *Ig* genes can be immunized. The mice produce high affinity fully human antibodies.
- Recombinant antibodies can be expressed on a large scale in plants.
- Combinatorial libraries of diabodies containing the H1 and H2 V_H CDR may be used to develop new drugs.

Affinity chromatography

- Insoluble immunoabsorbents prepared by coupling antibody to Sepharose can be used to affinity-purify antigens from complex mixtures and reciprocally to purify antibodies.

Neutralization of biological activity

- Antibodies can be detected by inhibition of biological functions such as viral infectivity or bacterial growth.
- Inhibition of biological function by known antibodies helps to define the role of the antigen, be it a hormone or cytokine for example, in complex responses *in vivo* and *in vitro*.

See the accompanying website (**www.roitt.com**) for multiple choice questions.